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Master Degree in Music and Acoustic Engineering
Numerical Modeling and Simulation in Acoustic

NMSA HWs

Report of the Homeworks

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Summary

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Homework 1

1.1 Weak Form and FE Discretization

In this chapter, the weak formulation and the corresponding finite element (FE) discretization of the boundary value problem introduced in Homework 1 are derived.

1.1.1 Strong Formulation

The one-dimensional wave propagation problem is considered:

$$\begin{cases} \rho(x)\omega^2 u(x) + \dfrac{\partial}{\partial x}\left(\mu(x)\dfrac{\partial u(x)}{\partial x}\right) = f \quad x \in (0, L) \\ \mu(0)\left.\dfrac{\partial u(x)}{\partial x}\right|_{x=0} = g_N \\ u(L) - \sqrt{\dfrac{\mu(L)}{\rho(L)}}\dfrac{i}{\omega}\left.\dfrac{\partial u(x)}{\partial x}\right|_{x=L} = g_A \end{cases} \tag{1.1}$$

In this report, the following notational simplifications are adopted:

- $u = u(x)$ and, analogously, for all functions depending on time;
- $u' = u'(x) = \frac{\partial u(x)}{\partial x}$ and, analogously, for all spatial derivatives;
- $u'(0) = \left.\frac{\partial u(x)}{\partial x}\right|_{x=0}$ and, analogously, for derivatives calculated in $x = L$.

The second boundary condition is an impedance (or absorbing) condition. It is useful to rewrite it in order to obtain $u'(L)$ explicitly. From the relation above, it follows that:

$$u'(L) = -i\omega\sqrt{\frac{\rho(L)}{\mu(L)}}\left(u(L) - g_A\right)$$

For convenience, the impedance coefficient is introduced as

$$\beta = \sqrt{\rho(L)\mu(L)}$$

1.1.2 Weak Formulation

To obtain the weak formulation, the strong form of the equation is multiplied by a test function $v(x) \in H^1(0, L)$ and integrated over $(0, L)$:

$$\int_0^L \rho\omega^2 uv dx + \int_0^L \frac{d}{dx}\left(\mu u'\right) v dx = \int_0^L f v dx$$

The second term is then integrated by parts:

$$\int_0^L \frac{d}{dx}(\mu u')v dx = [\mu u' v]_0^L - \int_0^L \mu u' v' dx$$

Substituting into the previous expression yields:

$$\omega^2 \int_0^L \rho u v dx - \int_0^L \mu u' v' dx + [\mu u' v]_0^L = \int_0^L f v dx$$

The boundary contributions are incorporated using the boundary conditions. At $x = 0$, it holds that:

$$\mu(0)u'(0)v(0) = g_N v(0) \quad \Longrightarrow \quad -\mu(0)u'(0)v(0) = -g_N v(0)$$

At $x = L$, using $u'(L)$ from the impedance condition, one obtains:

$$\mu(L)u'(L)v(L) = -i\omega\beta u(L)v(L) + i\omega\beta g_A v(L)$$

Collecting all terms, the weak formulation becomes:

$$\omega^2 \int_0^L \rho u v dx - \int_0^L \mu u' v' dx - i\omega\beta u(L)v(L) = \int_0^L f v dx + g_N v(0) - i\omega\beta g_A v(L) \tag{1.2}$$

It is convenient to identify the mass term, stiffness term and the contributions arising from the boundary conditions. The bilinear forms are defined as:

$$\mathcal{M}(u, v) = \int_0^L \rho u v dx, \quad \mathcal{A}(u, v) = \int_0^L \mu u' v' dx + i\omega\beta u(L)v(L)$$

and the linear functional as:

$$\mathcal{F}(v) = \int_0^L f v dx + g_N v(0) - i\omega\beta g_A v(L)$$

Thus, the weak formulation can be written compactly as:

$$\omega^2 \mathcal{M}(u, v) - \mathcal{A}(u, v) = \mathcal{F}(v) \quad \forall v \in H^1(0, L) \tag{1.3}$$

1.1.3 Finite Element Discretization and Algebraic System

A partition of the interval $(0, L)$ into N elements is introduced and the finite dimensional space $V_h \subset H^1(0, L)$, spanned by continuous, piecewise linear basis functions $\{\varphi_j\}_{j=1}^N$, is considered. The solution and the test function are approximated as:

$$u_h = u_h(x) = \sum_{j=1}^{N} u_j(x)\varphi_j(x), \quad v_h = v_h(x) = \varphi_i(x)$$

Substituting u_h and v_h into the equation (1.2) yields the discrete system. The entries of the mass matrix, stiffness matrix and load vector are then defined.

- Mass matrix:

$$M_{ij} = \int_0^L \varphi_i \varphi_j dx$$

- Stiffness matrix and impedance contribution:

$$A_{ij} = \int_0^L \mu\varphi_i'\varphi_j' dx + i\omega\beta\varphi_i(L)\varphi_j(L)$$

  Note that $\varphi_i(L) \neq 0$ is only for the basis function associated with the last node of the mesh. Therefore, the impedance term modifies only the entry A_{NN}.
- Load vector:

$$F_i = \int_0^L f\varphi_i dx + g_N\varphi_i(0) - i\omega\beta g_A\varphi_i(L)$$

  Here $\varphi_i(0)$ is nonzero only for the basis function associated with the first node, while $\varphi_i(L)$ is nonzero only for the last one.

Collecting all contributions, the linear system is obtained:

$$\omega^2\mathbf{M}\underline{u} - \mathbf{A}\underline{u} = \underline{F} \tag{1.4}$$

which corresponds to the discretized weak formulation and matches the form requested in the homework assignment. This completes the derivation of the weak formulation and its finite element discretization using linear basis functions.

1.2 MATLAB Solver

This section describes the MATLAB implementation used to solve the algebraic system derived in section 1.1 and introduces a verification procedure to assess the correctness of the finite element (FE) code. In particular, the Method of Manufactured Solutions (MMS) is adopted, which consists of prescribing an exact solution and deriving the corresponding forcing terms and boundary data. This provides a controlled test case where the numerical approximation can be directly compared against a known reference.

1.2.1 Verification Test Setup

Considering the one-dimensional domain and constant physical parameters :

$$\Omega = (0,1), \qquad L = 1, \qquad \rho = 1, \qquad \mu = 1.$$

For the verification test, the manufactured exact solution is imposed :

$$u_{\text{ex}}(x) = \sin(\omega x), \qquad \omega = 5\pi.$$

Internal forcing term Starting from the strong form in equation (1.1), the governing equation can be written as

$$\rho\omega^2 u(x) + \frac{d}{dx}\left(\mu u'(x)\right) = f(x) \tag{1.5}$$

where $u' = \frac{du}{dx}$. Computing the derivatives of the manufactured solution yields.

$$u_{\text{ex}}'(x) = \omega\cos(\omega x), \qquad u_{\text{ex}}''(x) = -\omega^2\sin(\omega x)$$

Since μ is constant, $\frac{d}{dx}(\mu u') = \mu u''$. Substituting u_{ex} into equation (1.5) with $\rho = \mu = 1$ yields

$$f(x) = \omega^2\sin(\omega x) + u_{\text{ex}}''(x) = \omega^2\sin(\omega x) - \omega^2\sin(\omega x) = 0$$

Therefore, the internal source term vanishes:

$$f(x) = 0 \tag{1.6}$$

Neumann condition The Neumann boundary condition in equation (1.1) at $x = 0$ reads

$$\mu(0)u'(0) = g_N \tag{1.7}$$

Using $u'_{\text{ex}}(x) = \omega\cos(\omega x)$ yields

$$u'_{\text{ex}}(0) = \omega\cos(0) = \omega, \qquad \mu(0) = 1$$

hence

$$g_N = \omega = 5\pi \tag{1.8}$$

Impedance condition The impedance boundary condition in equation (1.1) at $x = L$ is

$$u(L) - \sqrt{\frac{\mu(L)}{\rho(L)}}\frac{i}{\omega}u'(L) = g_A \tag{1.9}$$

Since $\rho(L) = \mu(L) = 1$, the impedance coefficient simplifies to

$$\sqrt{\frac{\mu(L)}{\rho(L)}} = 1$$

Evaluating the manufactured solution at $L = 1$ gives

$$u_{\text{ex}}(1) = \sin(5\pi) = 0, \qquad u'_{\text{ex}}(1) = 5\pi\cos(5\pi) = -5\pi$$

Substituting into the boundary condition yields

$$g_A = u_{\text{ex}}(1) - \frac{i}{\omega}u'_{\text{ex}}(1) = 0 - \frac{i}{5\pi}(-5\pi) = i$$

Therefore, the impedance boundary datum is

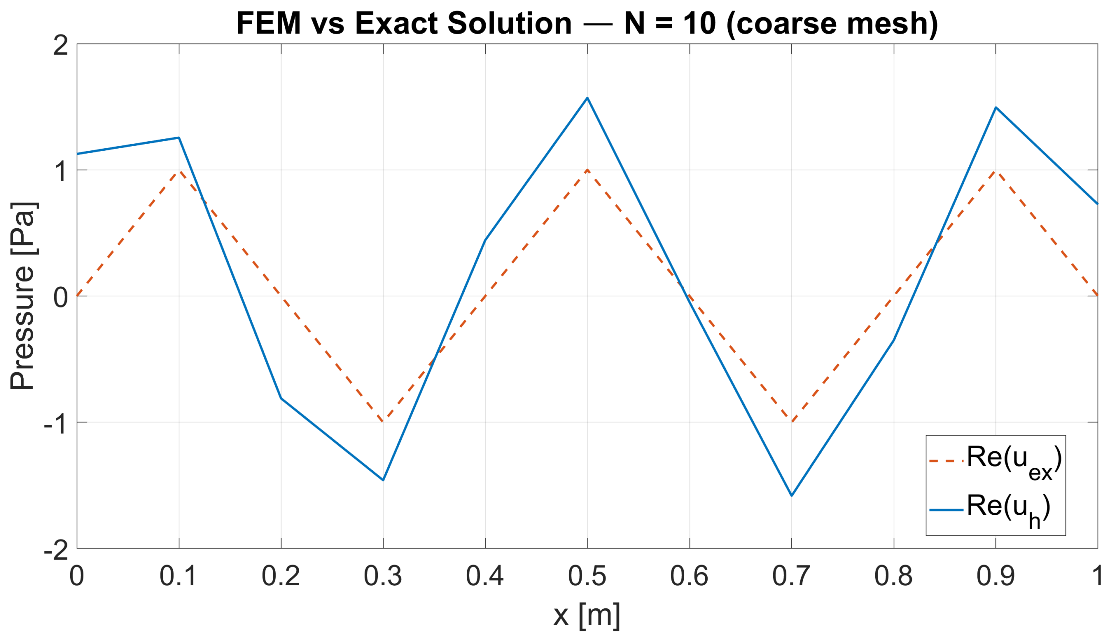
$$g_A = i \tag{1.10}$$

1.2.2 Numerical Results

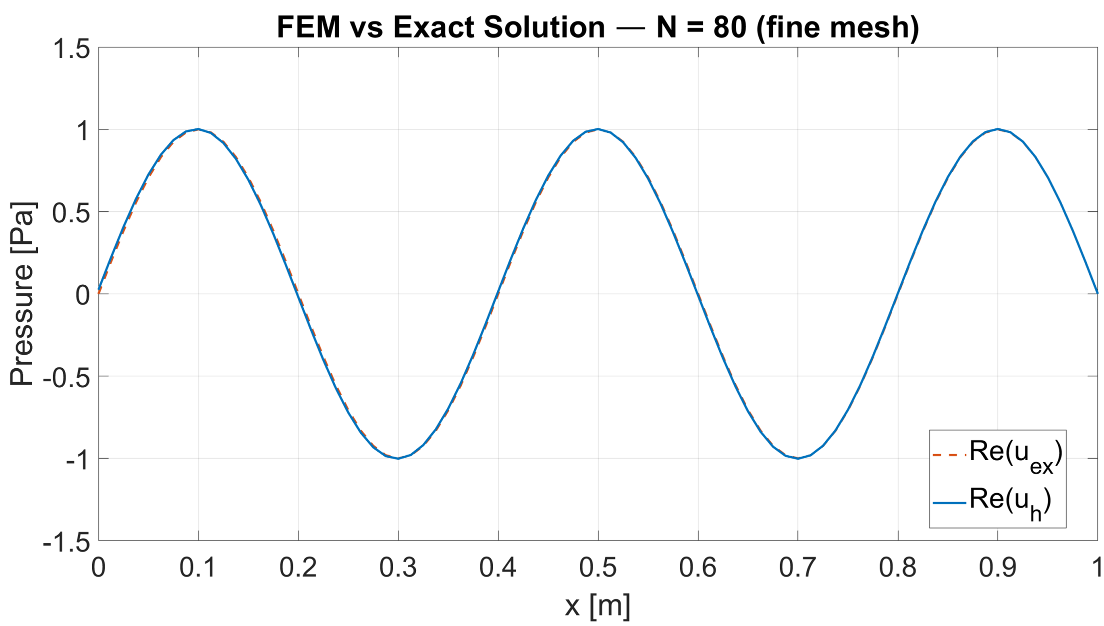
The algebraic system (1.4) is assembled and solved in MATLAB for a sequence of uniform meshes with $N \in \{10,\ 20,\ 40,\ 80\}$ elements. Since $\omega = 5\pi$, the wavelength of the exact solution is

$$\lambda = \frac{2\pi}{\omega} = \frac{2}{5} = 0.4\,\text{m}$$

so the domain $(0, 1)$ contains exactly 2.5 full wavelengths. On the coarsest mesh ($N = 10$, $h = 0.1$), each wavelength is covered by only $\lambda/h = 4$ elements, which is insufficient to resolve the oscillatory behavior of u_{ex}. On the finest mesh ($N = 80$, $h = 0.0125$), the resolution rises to 32 elements per wavelength, yielding an accurate approximation. Figure 1.1 compares the real part of the FE solution u_h with the exact solution u_{ex} on the coarse and fine meshes respectively. As expected, the coarse-mesh solution captures the correct frequency but with a noticeable phase and amplitude error, while the fine-mesh solution overlaps the exact curve almost perfectly.



(a) On the coarse mesh ($N = 10$, $h = 0.1$). The four elements per wavelength are insufficient to resolve the sinusoidal shape, resulting in a visible phase shift and amplitude overshoot



(b) On the fine mesh ($N = 80$, $h = 0.0125$). With 32 elements per wavelength the numerical and exact curves are virtually indistinguishable

Figure 1.1: Real part of the FE pressure field compared with the exact solution $u_{\text{ex}} = \sin(\omega x)$

1.2.3 Convergence Analysis

To quantify the approximation quality, the $L^2(0,1)$ error is computed for each mesh via numerical quadrature:

$$\|u - u_h\|_{L^2(0,L)} = \left(\int_0^L |u_{\text{ex}}(x) - u_h(x)|^2 dx \right)^{1/2}$$

The theoretical a-priori estimate for continuous piecewise-linear ($\mathbb{P}_1$) Galerkin approximations of elliptic problems guarantees :

$$\|u - u_h\|_{L^2(0,L)} \leq Ch^2 |u|_{H^2(0,L)} \tag{1.11}$$

so one expects a convergence rate of order 2 in the mesh size h. Table 1.1 reports the computed L^2 errors and the corresponding numerical convergence rates p, estimated from consecutive mesh pairs as

$$p = \frac{\log(e_i/e_{i+1})}{\log(h_i/h_{i+1})}$$

N	h	$\|u - u_h\|_{L^2}$	Rate p
10	0.1000	6.527×10^{-1}	—
20	0.0500	1.549×10^{-1}	2.07
40	0.0250	4.071×10^{-2}	1.93
80	0.0125	1.038×10^{-2}	1.97

Table 1.1: L^2 errors and estimated convergence rates for the MMS verification test with $u_{\text{ex}} = \sin(5\pi x)$ and NI boundary conditions.

The numerical rates confirm the expected $\mathcal{O}(h^2)$ convergence, consistent with the theoretical estimate (1.11) and with the use of $\mathbb{P}_1$ basis functions. The log-log convergence plot in figure 1.2 shows that the error decreases along a straight line of slope 2, parallel to the reference $\mathcal{O}(h^2)$ dashed line. This agreement between theory and numerics validates the correctness of the MATLAB implementation.

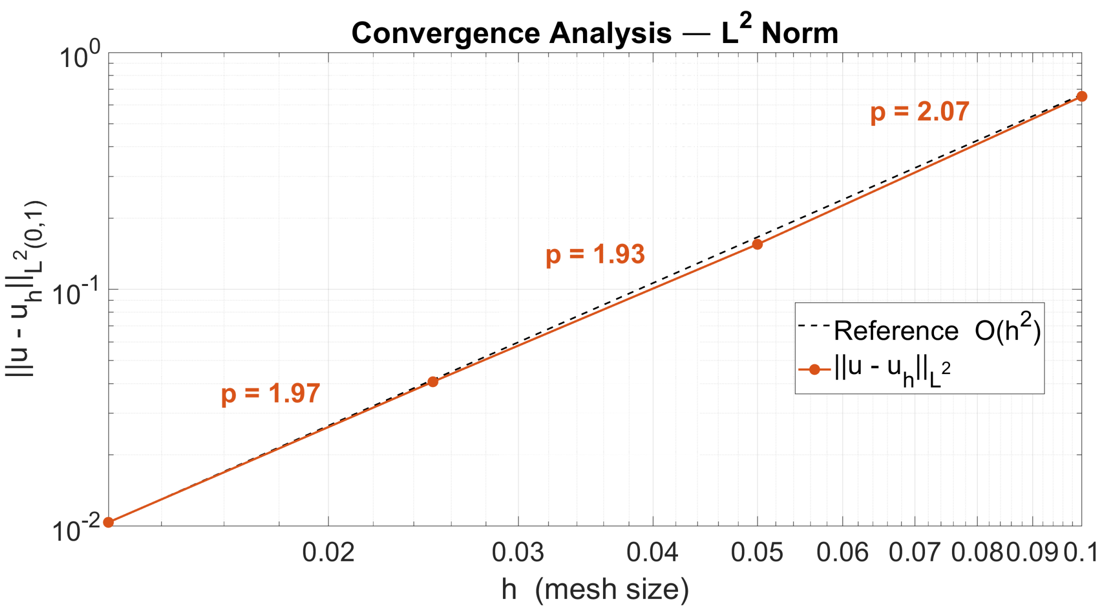


Figure 1.2: Log-log convergence plot of the L^2 error $\|u - u_h\|_{L^2(0,1)}$ as a function of the mesh size h for the MMS test with $u_{\text{ex}} = \sin(5\pi x)$. The dashed line represents the reference slope $\mathcal{O}(h^2)$ and the annotated rates confirm quadratic convergence for $\mathbb{P}_1$ elements

1.3 Wave Reflection at an Impedance Jump

The propagation of a time-harmonic field in a heterogeneous one-dimensional medium is considered, with a discontinuity in the material properties located at mid-domain. The main effect of such discontinuity is the generation of a partially reflected wave due to an impedance mismatch.

1.3.1 Problem Setup

The spatial domain is defined as $\Omega = (0,1)$ with constant density $\rho(x) = 1$. The stiffness is assumed piecewise constant and discontinuous at $x = 0.5$, according to

$$\mu(x) = \begin{cases} 4, & x \leq 0.5 \\ 1, & x > 0.5 \end{cases} \tag{1.12}$$

No volumetric source term is applied, hence $f(x) = 0$. The excitation is imposed at the left boundary through a Neumann condition $\mu(0)u'(0) = g_N$ with $g_N = 1$, whereas at the right boundary a perfectly absorbing condition is prescribed by setting $g_A = 0$. This choice prevents spurious reflections at $x = 1$, so that the observed reflected component is generated only at the internal interface $x = 0.5$.

1.3.2 Mechanical Impedance of the Two Media

For one-dimensional wave propagation, reflection and transmission at an interface are governed by the characteristic (mechanical) impedance of the medium, defined as

$$Z = \sqrt{\rho\mu} \tag{1.13}$$

Since $\rho = 1$ throughout Ω, the impedance depends only on μ. In the left subdomain $x \leq 0.5$ it follows that $Z_1 = \sqrt{1 \cdot 4} = 2$, while in the right subdomain $x > 0.5$ one obtains $Z_2 = \sqrt{1 \cdot 1} = 1$. The corresponding phase velocities are $c_1 = \sqrt{\mu_1/\rho} = 2$ and $c_2 = \sqrt{\mu_2/\rho} = 1$ respectively.

1.3.3 Reflection Coefficient at the Interface

When a right-going incident wave traveling in medium 1 reaches the interface with medium 2, continuity of displacement u and traction $\mu u'$ implies the formation of both a reflected and a transmitted component. The reflection coefficient for incidence from medium 1 to medium 2 is given by

$$R = \frac{Z_1 - Z_2}{Z_1 + Z_2} \tag{1.14}$$

Substituting $Z_1 = 2$ and $Z_2 = 1$ yields

$$R = \frac{2-1}{2+1} = \frac{1}{3}$$

Therefore, a partial reflection is expected at $x = 0.5$ with positive sign, i.e., without phase inversion.

1.3.4 Expected Qualitative Behavior in Amplitude and Phase

Distinct qualitative behaviors are expected on the two sides of the interface. In the left region $x < 0.5$, the superposition of the incident and reflected waves produces an interference pattern, corresponding to a partially standing-wave regime. As a consequence, oscillations are expected in both the amplitude profile $|u(x)|$ and the phase $\arg(u(x))$, which will be irregular and non-monotone. In the right region $x > 0.5$, the transmitted wave propagates toward $x = 1$. Due to the perfectly absorbing condition $g_A = 0$, no reflection is generated at the right boundary and the solution resembles a single traveling wave. Hence, the amplitude is expected to be comparatively smooth and the phase to grow linearly and monotonically with slope $\omega/c_2 = \omega$. At the interface $x = 0.5$, a discontinuity in the spatial derivative of the phase is expected, reflecting the change in phase velocity from $c_1 = 2$ to $c_2 = 1$: since the phase slope of a traveling wave is ω/c, the slope in the right subdomain is twice that in the left subdomain for a purely traveling wave.

1.3.5 Numerical Results

The problem is solved numerically using the MATLAB solver described in chapter 1.1, with $N = 200$ elements to ensure sufficient resolution across both subdomains. The stiffness coefficient is implemented as a piecewise function evaluated at the quadrature nodes of each element. Figure 1.3 reports the amplitude $|u(x)|$ and the unwrapped phase $\arg(u(x))$ of the computed solution.

1.3.6 Physical Interpretation

The numerical results are in full agreement with the analytical predictions derived in the previous sections. Three distinct regions can be identified in the solution.

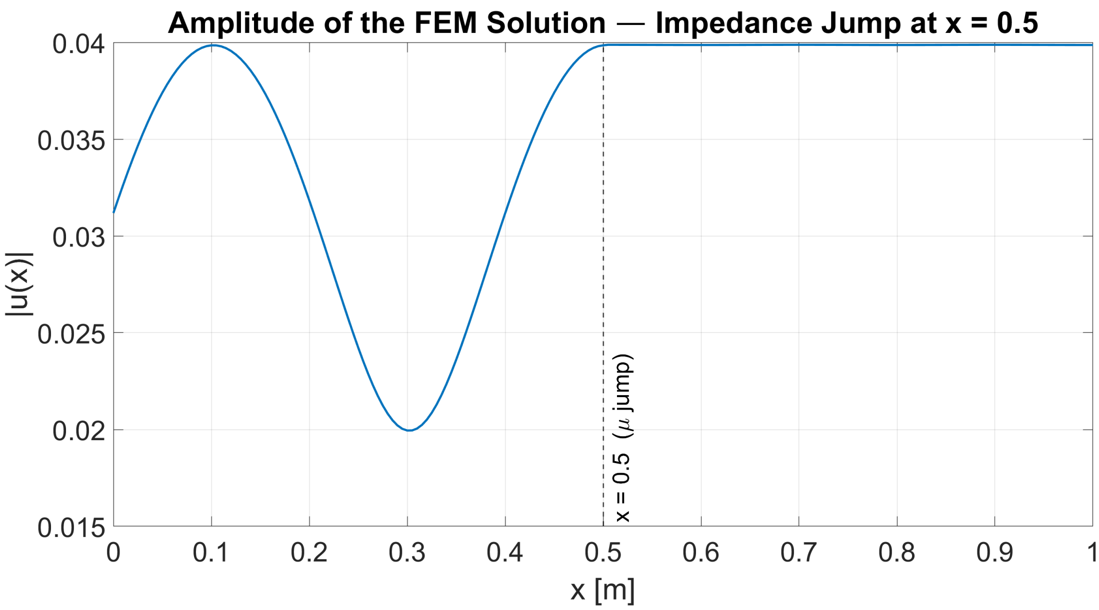
Left subdomain $x < 0.5$: partial standing wave The amplitude profile $|u(x)|$ exhibits oscillations with alternating local maxima and minima, which are the signature of the interference between the rightward-traveling incident wave and the leftward-traveling reflected wave generated at the interface. The phase is irregular and non-monotone, consistently with the superposition of two counter-propagating components. The reflection coefficient $R = 1/3$ implies that approximately 11% of the incident energy is reflected back into the left subdomain, producing a moderately pronounced standing-wave pattern.

Right subdomain $x > 0.5$: traveling wave Beyond the interface, the amplitude is smooth and the phase grows linearly with slope $\omega/c_2 = \omega$, confirming the presence of a single right-going traveling wave. The absorbing boundary condition at $x = 1$ effectively prevents any reflection from the right end, so no additional interference pattern appears in this region.

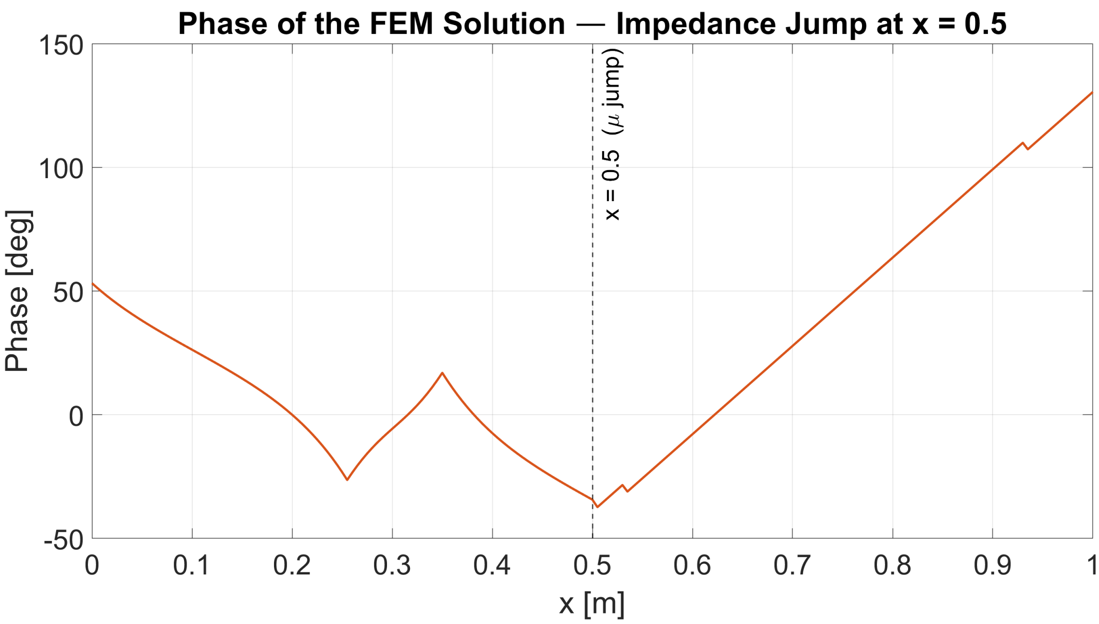
Interface $x = 0.5$: phase velocity change A clear kink in the phase is visible at the interface. Since the phase slope of a traveling wave equals ω/c, the ratio of the slopes on the two sides of the interface is

$$\frac{(\omega/c_2)}{(\omega/c_1)} = \frac{c_1}{c_2} = \frac{2}{1} = 2$$

confirming that the phase advances twice as fast per unit length in the right medium, consistently with $c_2 = 1 < c_1 = 2$.



(a) Amplitude $|u(x)|$ of the numerical solution. The oscillatory pattern in the left subdomain ($x < 0.5$) is the signature of the standing-wave component generated by the partial reflection at the interface, while the smoother decay in the right subdomain ($x > 0.5$) corresponds to the purely transmitted wave



(b) Unwrapped phase arg$(u(x))$ of the numerical solution. The irregular, non-monotone behavior for $x < 0.5$ is due to the interference between the incident and reflected waves, while the linear monotone growth for $x > 0.5$ is consistent with a single right-going traveling wave. The kink at $x = 0.5$ reflects the change in phase velocity between the two media

Figure 1.3: Amplitude and unwrapped phase of the FE solution for the impedance-jump problem with μ as in equation (1.12), $\rho = 1$, $g_N = 1$, $g_A = 0$ and $\omega = 5\pi$

1.4 Numerical Dispersion Analysis

It is well known that spatial discretization introduces a dispersion error: the discrete solution does not propagate with the same phase velocity as the continuous (physical) wave. In the frequency domain, this phenomenon is conveniently quantified by comparing the continuous wavenumber k with the discrete (numerical) wavenumber k_h, associated with the finite-dimensional approximation. The resulting discrepancy represents the numerical dispersion of the method.

1.4.1 Eigenvalue Problem Setting

To isolate dispersion effects from external excitations, the homogeneous problem is considered by setting $f = 0$ and imposing homogeneous boundary data. In particular, a homogeneous Neumann condition is prescribed at the left boundary, namely $g_N = 0$, which implies $\mu u'(0) = 0$, while a homogeneous Dirichlet condition is imposed at the right boundary, $u(L) = 0$. Under these assumptions, the Helmholtz model reduces to an eigenvalue problem for the spatial operator.

1.4.2 Continuous and Discrete Formulations

In the continuous setting and for constant coefficients $\rho = \mu = 1$, the eigenmodes satisfy the classical second-order equation

$$-u''(x) = k^2 u(x) \tag{1.15}$$

where k^2 denotes the exact eigenvalues of the continuous problem and k is the corresponding continuous wavenumber. By applying the finite element discretization derived in section 1.1 and restricting the solution to the discrete space, the homogeneous algebraic system assumes the form of a generalized eigenvalue problem. Denoting by A the stiffness matrix and by M the mass matrix, the discrete modes satisfy

$$\mathbf{A}\underline{u} = k_h^2 \mathbf{M}\underline{u} \tag{1.16}$$

where k_h^2 are the discrete eigenvalues and k_h is the associated numerical wavenumber. The dispersion error is therefore interpreted as the difference between the spectra of the continuous and discrete operators, expressed in terms of wavenumbers.

1.4.3 Numerical Results

The accuracy of the method is assessed by introducing the dimensionless resolution parameter h/λ, where h denotes the characteristic mesh size and λ the wavelength. The numerical dispersion is represented through the dispersion curve, defined as the ratio k_h/k plotted as a function of h/λ. The generalized eigenvalue problem (1.16) is solved in MATLAB for a sequence of meshes with $N \in \{10,\ 20,\ 40,\ 80,\ 160\}$ elements. For each discrete eigenvalue k_h^2, the corresponding exact wavenumber k is identified analytically and the ratio k_h/k is computed. The resulting dispersion curve is reported in figure 1.4, together with the asymptotic Taylor expansion

$$\frac{k_h}{k} \approx 1 + \frac{\pi^2}{6}\left(\frac{h}{\lambda}\right)^2 \tag{1.17}$$

which holds for the consistent-mass $\mathbb{P}_1$ FEM in the well-resolved regime $h/\lambda \ll 1$.

1.4.4 Physical Interpretation

The dispersion curve in figure 1.4 reveals the characteristic overestimation behavior of the consistent-mass $\mathbb{P}_1$ FEM: unlike lumped-mass formulations, which underestimate the physical wavenumber, the consistent-mass scheme yields $k_h > k$, so that the ratio k_h/k is strictly greater than unity throughout the resolved spectrum. Three distinct resolution regimes can be identified.

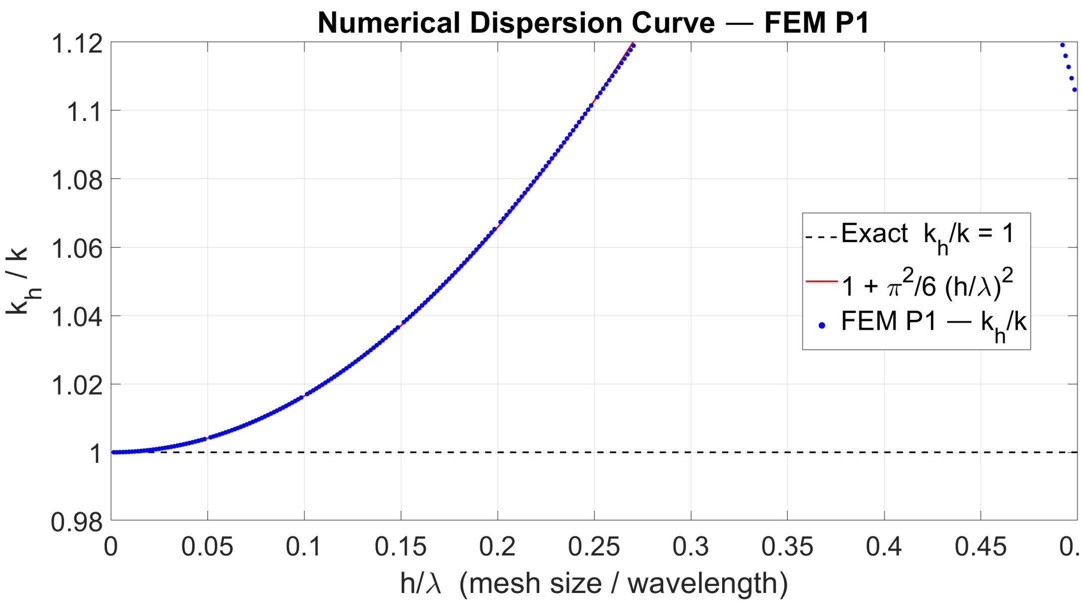


Figure 1.4: Numerical dispersion curve k_h/k as a function of the resolution parameter h/λ for the consistent-mass $\mathbb{P}_1$ FEM (blue dots). The dashed black line at $k_h/k = 1$ is the exact non-dispersive reference, while the solid parabola represents the asymptotic Taylor expansion $1 + \frac{\pi^2}{6}(h/\lambda)^2$. For well-resolved meshes the two curves coincide, whereas near the Nyquist limit the exact discrete symbol departs from the parabola, reaches a local maximum and settles at approximately $k_h/k \approx 1.10$ at $h/\lambda = 0.5$

Fine meshes: $h/\lambda \to 0$ In the well-resolved regime, the numerical results follow the asymptotic parabola (1.17) with remarkable accuracy, confirming the theoretical second-order convergence of the dispersion error. The overestimation of the wavenumber is practically negligible: below $h/\lambda \approx 0.05$, corresponding to 20 or more elements per wavelength, the deviation from unity is smaller than 0.1% and the numerical phase velocity coincides with the physical one to engineering accuracy.

Intermediate resolution At $h/\lambda = 0.1$, i.e. approximately 10 elements per wavelength, the ratio $k_h/k \approx 1.017$, corresponding to a dispersion error of about 1.7%. This level of resolution is commonly considered a practical minimum for engineering applications where moderate accuracy is acceptable. The asymptotic expansion (1.17) remains in close agreement with the exact discrete symbol up to this range.

Coarse meshes: approach to the Nyquist limit As h/λ increases beyond 0.2, the exact dispersion curve progressively departs from the parabolic approximation, which diverges monotonically, whereas the physical discrete symbol reaches a local maximum and subsequently decreases. Near the Nyquist limit $h/\lambda \to 0.5$, the curve settles at a finite value of approximately $k_h/k \approx 1.10$. This behavior, captured in full by the numerical computation since the MATLAB solver resolves the complete discrete spectrum without restriction, demonstrates that the parabolic expansion is a low-frequency asymptotic and does not bound the solution at high wavenumbers. Points in the plot that momentarily exceed the upper axis limit before returning to $k_h/k \approx 1.10$ are therefore physically meaningful and not artifacts: they correspond to the exact symbol of the discrete operator evaluated in the poorly resolved regime.

General rule of thumb The analysis confirms that at least 10 elements per wavelength ($h/\lambda \leq 0.1$) are required to keep the dispersion error below 2% and that 20 or more elements per wavelength ($h/\lambda \leq 0.05$) are needed for high-fidelity simulations. These thresholds are consistent with the standard engineering guidelines for $\mathbb{P}_1$ FEM applied to wave propagation problems.

1.5 Perfectly Matched Layer (PML)

In order to reduce spurious reflections at the right boundary, an alternative to the absorbing (impedance) boundary condition at $x = 1$ is considered. The computational domain is extended by adding an absorbing layer, referred to as a Perfectly Matched Layer (PML), in the region $x \in (1, 1 + L_{\text{PML}})$. The PML is designed to damp outgoing waves without reflecting them back into the physical domain $\Omega = (0, 1)$.

1.5.1 Damping Profile

The attenuation mechanism is introduced through a spatially varying damping function $\sigma(x)$. In order to achieve effective absorption while minimizing numerical reflections due to abrupt coefficient variations, $\sigma(x)$ is required to increase smoothly from zero at the interface $x = 1$. A parabolic profile is adopted, defined as

$$\sigma(x) = \begin{cases} 0 & \text{for } x \leq 1 \\ \sigma_{\max} \left(\dfrac{x - 1}{L_{\text{PML}}} \right)^2 & \text{for } x > 1 \end{cases} \tag{1.18}$$

1.5.2 Complex Stretching and Modified Coefficients

The PML formulation is obtained by introducing the complex stretching function

$$s(x) = 1 - i\frac{\sigma(x)}{\omega} \tag{1.19}$$

Within the layer, the Helmholtz model is recast by replacing the physical parameters ρ and μ with complex-valued equivalents,

$$\tilde{\rho}(x) = \rho(x)s(x), \qquad \tilde{\mu}(x) = \frac{\mu(x)}{s(x)} \tag{1.20}$$

This choice preserves the structure of the governing equation while introducing attenuation through complex coefficients.

1.5.3 Impedance Matching and Exponential Decay

A key property of the PML is that it is perfectly matched at the interface $x = 1$. In particular, the characteristic impedance in the PML region satisfies

$$Z_{\text{PML}} = \sqrt{\tilde{\rho}(x)\tilde{\mu}(x)} = \sqrt{\rho(x)s(x)\frac{\mu(x)}{s(x)}} = \sqrt{\rho(x)\mu(x)} = Z \tag{1.21}$$

which shows that the impedance is continuous across the interface. This condition implies, at the continuous level, perfect transmission and therefore the absence of reflected waves at

$x = 1$. On the other hand, the local wavenumber becomes complex. Defining $k = \omega\sqrt{\rho/\mu}$ as the physical wavenumber, the PML wavenumber is obtained as

$$k_{\text{PML}}(x) = \omega\sqrt{\frac{\tilde{\rho}(x)}{\tilde{\mu}(x)}} = \omega\sqrt{\frac{\rho(x)s(x)}{\mu(x)/s(x)}} = \omega\sqrt{\frac{\rho(x)}{\mu(x)}}s(x) = ks(x) \tag{1.22}$$

Since $s(x) = 1 - i\sigma(x)/\omega$ has a negative imaginary part for $\sigma(x) \geq 0$, the transmitted wave in the layer experiences an exponential decay in amplitude as x increases, leading to strong attenuation before reaching the truncated boundary.

1.5.4 Boundary Conditions on the Extended Domain

The excitation at the left boundary is kept unchanged and the Neumann condition is imposed at $x = 0$. The right boundary is moved to the end of the PML, located at $x = 1 + L_{\text{PML}}$, where a homogeneous Dirichlet condition is prescribed, $u(1+L_{\text{PML}}) = 0$. This truncation is justified by the assumption that the outgoing wave has been sufficiently damped by the PML and is therefore negligible at the outer boundary.

1.5.5 Numerical Results

The PML approach is validated by comparing its solution against the impedance boundary condition result obtained in section 1.3, for the same physical setup: $\mu(x)$ as in equation (1.12), $\rho = 1$, $g_N = 1$ and $\omega = 5\pi$. The computational domain is extended to $(0,\ 1.2)$ with $L_{\text{PML}} = 0.2$ and $\sigma_{\text{max}} = 100$, discretized uniformly with a total of $N = 240$ elements. Figure 1.5 reports the amplitude $|u(x)|$ and the unwrapped phase $\arg(u(x))$ over the entire extended domain $(0,\ 1.2)$, allowing the behavior inside the absorbing layer to be observed directly alongside the reference impedance BC solution restricted to $(0, 1)$.

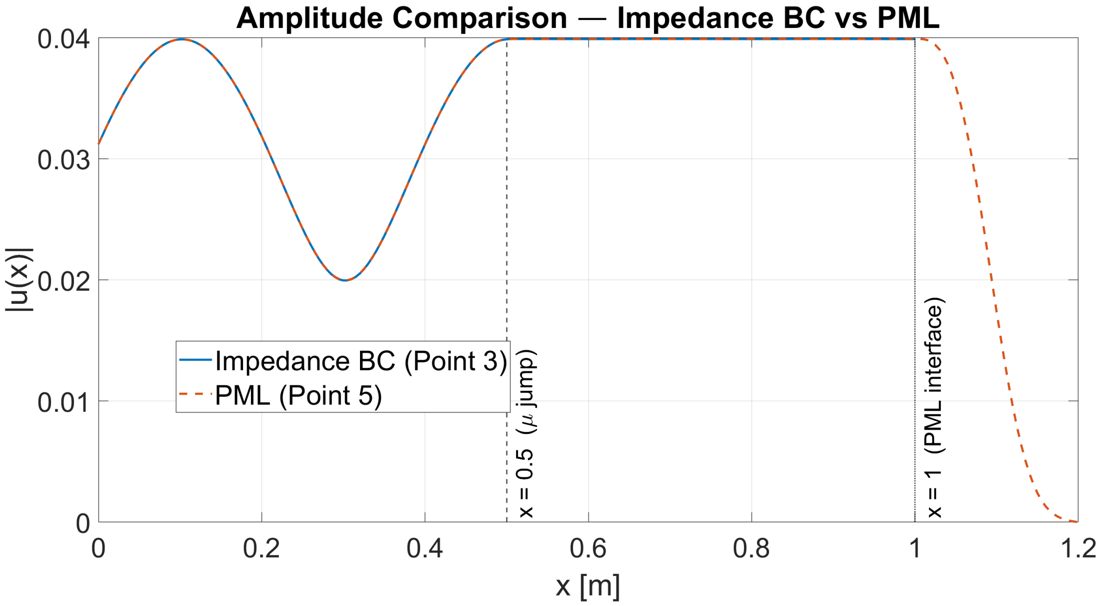
1.5.6 Physical Interpretation

The results in figure 1.5 demonstrate the two distinct roles played by the PML: it acts as a transparent boundary for the physical domain while simultaneously providing strong attenuation of the transmitted field in the layer region.

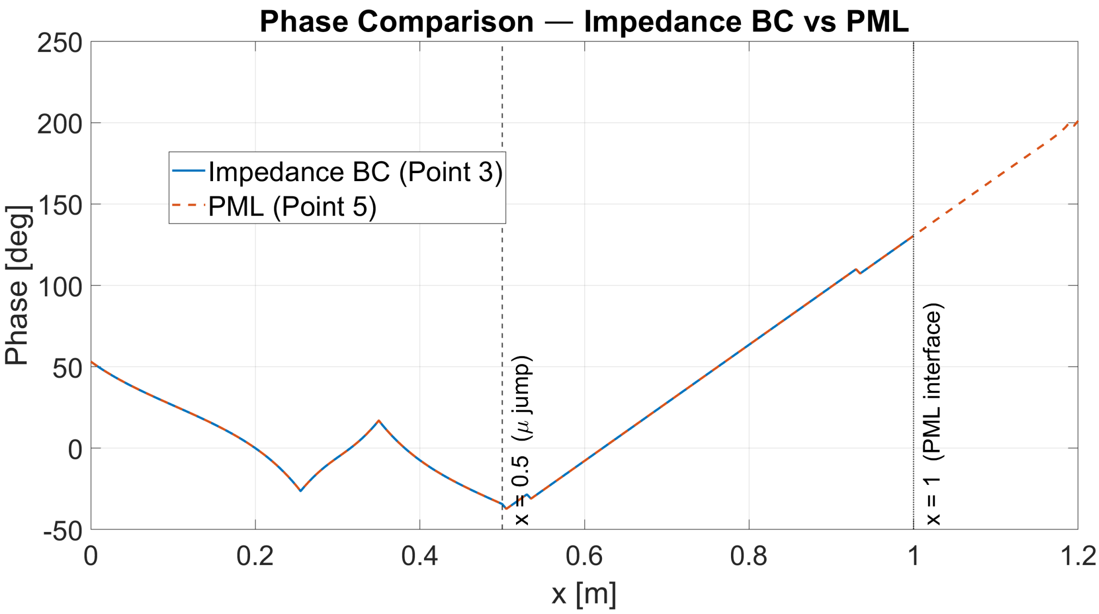
Physical domain $0 \leq x \leq 1$: perfect agreement The amplitude and phase of the PML solution overlap exactly with those of the impedance BC reference throughout the entire physical domain. The oscillatory interference pattern for $x < 0.5$ and the smooth transmitted-wave profile for $x > 0.5$ are reproduced identically by both approaches. This confirms that the impedance-matching property (1.21) is effectively realized at the discrete level, and that no spurious reflection is introduced at the interface $x = 1$.

Absorbing layer $1 < x \leq 1.2$: exponential decay Within the PML, the amplitude decays exponentially and reaches values negligibly close to zero well before the outer boundary at $x = 1.2$, validating the effectiveness of the parabolic damping profile (1.18) with $\sigma_{\text{max}} = 100$. This rapid attenuation justifies the application of the homogeneous Dirichlet condition at the truncation point without introducing significant error into the physical domain.

Absorbing layer $1 < x \leq 1.2$: linear phase decay While the amplitude vanishes, the unwrapped phase of the PML solution continues to evolve linearly and monotonically inside



(a) Amplitude $|u(x)|$. In the physical domain $(0, 1)$ the two curves are virtually indistinguishable, confirming the absence of spurious reflections at the interface $x = 1$. In the absorbing layer $(1,\ 1.2)$ the PML solution exhibits rapid exponential decay toward zero, demonstrating the effectiveness of the complex damping coefficient (1.20)



(b) Unwrapped phase arg$(u(x))$. In the physical domain $(0, 1)$ the phase profiles of the impedance BC and PML solutions coincide exactly. Inside the absorbing layer $(1,\ 1.2)$ the phase of the PML solution continues to decrease linearly, a signature of an outgoing traveling wave whose amplitude is simultaneously being attenuated by the complex stretching function (1.19)

Figure 1.5: Amplitude $|u(x)|$ and unwrapped phase arg$(u(x))$ for the impedance BC solution (Point 3, section 1.3) and the PML solution (Point 5, section 1.5) over the extended domain $(0,\ 1.2)$. The dashed vertical line marks the physical–PML interface at $x = 1$

the layer. This behavior is physically crucial: it demonstrates that the wave inside the PML retains the character of an outgoing traveling wave, propagating in the positive x direction with the same phase velocity as in the physical domain, while its energy content is simultaneously dissipated by the imaginary part of the complex stretching function $s(x)$. The coexistence of phase propagation and amplitude attenuation is precisely the defining property of the PML, which distinguishes it from simpler damping strategies that also distort the phase and can therefore introduce reflections.

2

Homework 2

2.1 Equivalence to the Second-Order Wave Equation

The linearized shallow water equations govern the propagation of surface waves in a shallow domain $(x,t) \in (0,L) \times (0,T)$. The equivalence between the first-order coupled system and the classical second-order wave equation is established by eliminating one of the two unknowns.

2.1.1 Strong Formulation

The governing system reads:

$$\begin{cases} \eta_t(x,t) + q_x(x,t) = 0 \\ q_t(x,t) + gH\eta_x(x,t) = 0 \end{cases} \tag{2.1}$$

where $\eta(x,t)$ is the free surface elevation, $q(x,t)$ is the discharge, H is the mean water depth, and g is the gravitational acceleration. The system is supplemented by periodic boundary conditions:

$$\eta(0,t) = \eta(L,t), \quad q(0,t) = q(L,t) \tag{2.2}$$

2.1.2 Decoupling the System

The discharge variable $q(x,t)$ is eliminated by cross-differentiation. Differentiating the first equation of (2.1) with respect to time t:

$$\eta_{tt}(x,t) + q_{xt}(x,t) = 0 \quad \Longrightarrow \quad \eta_{tt}(x,t) = -q_{xt}(x,t) \tag{2.3}$$

Differentiating the second equation of (2.1) with respect to space x:

$$q_{tx}(x,t) + gH\eta_{xx}(x,t) = 0 \quad \Longrightarrow \quad q_{tx}(x,t) = -gH\eta_{xx}(x,t) \tag{2.4}$$

Assuming sufficient regularity, specifically $q \in C^2((0,L) \times (0,T))$, Schwarz's theorem yields $q_{xt} = q_{tx}$. Substituting (2.4) into (2.3) results in the standard second-order wave equation:

$$\eta_{tt}(x,t) = gH\eta_{xx}(x,t) \quad \text{for } (x,t) \in (0,L) \times (0,T) \tag{2.5}$$

2.1.3 Determination of Suitable Boundary Conditions

The periodic boundary conditions on the coupled variables (η, q) must be translated into conditions solely on η and its derivatives. The first condition is already explicit:

$$\eta(0,t) = \eta(L,t) \tag{2.6}$$

To convert the second condition $q(0,t) = q(L,t)$, both sides are differentiated with respect to t, yielding $q_t(0,t) = q_t(L,t)$. From the second equation of (2.1), $q_t = -gH\eta_x$. Evaluating at both boundaries and substituting:

$$-gH\eta_x(0,t) = -gH\eta_x(L,t) \quad \Longrightarrow \quad \eta_x(0,t) = \eta_x(L,t) \tag{2.7}$$

2.1.4 Equivalence

The first-order system (2.1) is therefore fully equivalent to the second-order acoustic wave equation:

$$\eta_{tt}(x,t) = gH\eta_{xx}(x,t), \quad (x,t) \in (0,L) \times (0,T) \tag{2.8}$$

subject to the periodic boundary conditions $\eta(0,t) = \eta(L,t)$ and $\eta_x(0,t) = \eta_x(L,t)$, derived in (2.6) and (2.7) respectively.

2.2 Conservation of Energy

The conservation of mechanical energy under periodic boundary conditions is established by showing that the time derivative of the prescribed energy functional vanishes identically.

2.2.1 Definition of the Energy Functional

The energy functional of the shallow water system over the spatial domain $(0, L)$ is defined as:

$$E(t) = \int_0^L \left(\frac{1}{2} g\eta^2 + \frac{1}{2}\frac{q^2}{H} \right) dx \tag{2.9}$$

where the first term represents the potential energy associated with the free surface elevation η, and the second term represents the kinetic energy associated with the discharge q. To prove conservation, it suffices to show that $\frac{dE}{dt} = 0$.

2.2.2 Time Derivative of the Energy

Assuming sufficient smoothness of $\eta(x,t)$ and $q(x,t)$, the Leibniz rule permits the exchange of differentiation and integration:

$$\frac{dE}{dt} = \frac{d}{dt}\int_0^L \left(\frac{1}{2} g\eta^2 + \frac{1}{2}\frac{q^2}{H} \right) dx = \int_0^L \frac{\partial}{\partial t}\left(\frac{1}{2} g\eta^2 + \frac{1}{2}\frac{q^2}{H} \right) dx \tag{2.10}$$

Applying the chain rule to compute the partial time derivatives of the integrand yields:

$$\frac{dE}{dt} = \int_0^L \left(g\eta\eta_t + \frac{1}{H} q q_t \right) dx \tag{2.11}$$

2.2.3 Substitution of the Strong Formulation

Recalling the linearized shallow water equations (2.1):

$$\begin{cases} \eta_t = -q_x \\ q_t = -gH\eta_x \end{cases} \tag{2.12}$$

Substituting these expressions for η_t and q_t into (2.11):

$$\frac{dE}{dt} = \int_0^L \left(g\eta(-q_x) + \frac{1}{H} q(-gH\eta_x) \right) dx \tag{2.13}$$

2.2.4 Algebraic Simplification and the Product Rule

Factoring out the common constant $-g$, the integral simplifies to:

$$\frac{dE}{dt} = -g \int_0^L (\eta q_x + q\eta_x)\, dx \tag{2.14}$$

The integrand is recognized as the exact spatial derivative of the product ηq via the product rule $\frac{\partial}{\partial x}(\eta q) = \eta_x q + \eta q_x$. Therefore:

$$\frac{dE}{dt} = -g \int_0^L \frac{\partial}{\partial x}(\eta q) dx \tag{2.15}$$

2.2.5 Application of the Fundamental Theorem of Calculus

Integrating the exact derivative over $[0, L]$ yields the evaluation of the boundary terms:

$$\frac{dE}{dt} = -g \Big[\eta(x,t) q(x,t) \Big]_{x=0}^{x=L} = -g \left(\eta(L,t)q(L,t) - \eta(0,t)q(0,t) \right) \tag{2.16}$$

2.2.6 Application of Boundary Conditions

The periodic boundary conditions prescribe $\eta(0,t) = \eta(L,t)$ and $q(0,t) = q(L,t)$. Substituting these into the boundary evaluation:

$$\eta(L,t)q(L,t) - \eta(0,t)q(0,t) = \eta(L,t)q(L,t) - \eta(L,t)q(L,t) = 0 \tag{2.17}$$

2.2.7 Energy Conservation

Therefore, the time derivative of the energy is exactly zero:

$$\frac{dE}{dt} = 0 \tag{2.18}$$

This rigorously demonstrates that the energy $E(t)$ defined in (2.9) is conserved for all $t \in (0, T)$ under the prescribed periodic boundary conditions.

2.3 Weak Form and Spectral Element Discretization

A variational formulation of the linearized shallow water system is derived and subsequently discretized using the Spectral Element Method (SEM), leading to a global first-order block ODE system suitable for time integration.

2.3.1 Function Spaces

The appropriate function space is defined to accommodate the periodic boundary conditions $\eta(0,t) = \eta(L,t)$ and $q(0,t) = q(L,t)$. Solutions are sought in the space of periodic functions with square-integrable first derivatives:

$$V = H^1_{\text{per}}(0, L) = \{v \in H^1(0, L) : v(0) = v(L)\}$$

For any $t \in (0, T)$, it is assumed that $\eta(\cdot, t) \in V$ and $q(\cdot, t) \in V$. Test functions are drawn from the same space: $v, w \in V$.

2.3.2 Weak Formulation

Starting from the strong form of the system:

$$\begin{cases} \eta_t + q_x = 0 \\ q_t + gH\eta_x = 0 \end{cases} \tag{2.19}$$

each equation is multiplied by a test function and integrated over the domain $(0, L)$:

$$\int_0^L \eta_t v dx + \int_0^L q_x v dx = 0 \quad \forall v \in V$$

$$\int_0^L q_t w dx + gH \int_0^L \eta_x w dx = 0 \quad \forall w \in V$$

Boundary Treatment Applying integration by parts to the convective term in the first equation:

$$\int_0^L q_x v dx = [qv]_0^L - \int_0^L q v_x dx$$

Since both q and v belong to V, the boundary term vanishes exactly by the periodicity constraint:

$$[qv]_0^L = q(L)v(L) - q(0)v(0) = 0$$

For the continuous Galerkin SEM with periodic basis functions, it is standard to retain the spatial derivatives on the trial functions. The weak formulation therefore reads: find $\eta, q \in V$ such that for all $v, w \in V$,

$$\begin{cases} \int_0^L \eta_t v dx + \int_0^L q_x v dx = 0 \\ \int_0^L q_t w dx + gH \int_0^L \eta_x w dx = 0 \end{cases} \tag{2.20}$$

2.3.3 Spectral Element Discretization

Mesh and Element Partition The domain $\Omega = (0, L)$ is partitioned into N_e non-overlapping elements $\Omega_e = [x_{e-1}, x_e]$. Each element is mapped to the reference element $[-1, 1]$ via an affine transformation.

Basis Functions A finite-dimensional subspace $V_h \subset V$ is introduced. In SEM, the basis functions $\phi_j(x)$ are piecewise continuous polynomials of degree N, specifically Lagrange polynomials constructed over Gauss-Lobatto-Legendre (GLL) nodes. The continuous solutions are approximated as:

$$\eta(x,t) \approx \eta_h(x,t) = \sum_{j=1}^{N_{\text{dof}}} \eta_j(t)\phi_j(x), \qquad q(x,t) \approx q_h(x,t) = \sum_{j=1}^{N_{\text{dof}}} q_j(t)\phi_j(x)$$

where N_{dof} is the total number of global degrees of freedom, with the periodic assembly mapping the last node to the first.

2.3.4 Galerkin System and Matrix Assembly

Substituting the discrete approximations and choosing $v = w = \phi_i(x)$, the semi-discrete system becomes:

$$\sum_j \dot{\eta}_j \int_0^L \phi_j\phi_i dx + \sum_j q_j \int_0^L \phi_j'\phi_i dx = 0$$

$$\sum_j \dot{q}_j \int_0^L \phi_j\phi_i dx + gH \sum_j \eta_j \int_0^L \phi_j'\phi_i dx = 0$$

The global mass matrix $\mathbf{M}$ and the first-derivative matrix $\mathbf{D}$ are defined by:

$$M_{ij} = \int_0^L \phi_i\phi_j dx, \qquad D_{ij} = \int_0^L \phi_i\phi_j' dx$$

Note on GLL Quadrature In SEM, integrals are evaluated using GLL quadrature. The discrete orthogonality property of the Lagrange basis at the GLL nodes renders $\mathbf{M}$ diagonal (lumped), which is advantageous for time integration. Introducing the global coefficient vectors:

$$\underline{\eta}(t) = \begin{bmatrix} \eta_1(t) \\ \vdots \\ \eta_{N_{\text{dof}}}(t) \end{bmatrix}, \qquad \underline{q}(t) = \begin{bmatrix} q_1(t) \\ \vdots \\ q_{N_{\text{dof}}}(t) \end{bmatrix}$$

the semi-discrete system takes the compact matrix form:

$$\mathbf{M}\dot{\underline{\eta}}(t) + \mathbf{D}\underline{q}(t) = \underline{0} \tag{2.21}$$

$$\mathbf{M}\dot{\underline{q}}(t) + gH\mathbf{D}\underline{\eta}(t) = \underline{0} \tag{2.22}$$

2.3.5 Final Block System

Stacking the unknown vectors into the global state vector $\underline{U} = \begin{bmatrix} \underline{\eta} \\ \underline{q} \end{bmatrix}$, the coupled system (2.21)-(2.22) is cast into the block matrix form:

$$\begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{M} \end{bmatrix} \begin{bmatrix} \dot{\underline{\eta}} \\ \dot{\underline{q}} \end{bmatrix} = - \begin{bmatrix} \mathbf{0} & \mathbf{D} \\ gH\mathbf{D} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \underline{\eta} \\ \underline{q} \end{bmatrix} \tag{2.23}$$

Defining the global block matrices:

$$\mathcal{M} = \begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{M} \end{bmatrix}, \qquad \mathcal{S} = \begin{bmatrix} \mathbf{0} & \mathbf{D} \\ gH\mathbf{D} & \mathbf{0} \end{bmatrix} \tag{2.24}$$

the target semi-discrete first-order ODE system is recovered:

$$\mathcal{M}\dot{\underline{U}} + \mathcal{S}\underline{U} = \mathbf{0} \tag{2.25}$$

Here, $\mathcal{M}$ is symmetric and positive definite (diagonal under GLL quadrature), while $\mathcal{S}$ encodes the anti-symmetric spatial derivative operator scaled by the physical parameters.

2.4 MATLAB Solver

The semi-discrete block system derived in section 2.3 is advanced in time by means of the Crank-Nicolson scheme. The implementation is validated against the exact traveling-wave solution associated with the Gaussian initial data prescribed by the homework assignment. Accuracy is assessed through direct solution snapshots, h-refinement and p-enrichment convergence studies.

2.4.1 Numerical Setup and Time Integration

The numerical experiments are performed on the interval $(0, L)$ with

$$L = 1\,\mathrm{m}, \qquad H = 1\,\mathrm{m}, \qquad g = 9.81\,\mathrm{m/s^2}$$

yielding the wave speed

$$c = \sqrt{gH} = \sqrt{9.81}\,\mathrm{m/s} \tag{2.26}$$

and a final time $T = 0.5\,\mathrm{s}$. The initial conditions are the Gaussian pulse prescribed by the assignment:

$$\eta_0(x) = e^{-50(x-0.5)^2}, \qquad q_0(x) = ce^{-50(x-0.5)^2} \tag{2.27}$$

Since the datum (2.27) satisfies the compatibility condition $q_0 = c\eta_0$, the exact solution is the periodic right-traveling wave

$$\eta_{\text{ex}}(x,t) = \exp\left[-50\left(\text{mod}(x-ct, L) - 0.5\right)^2\right] \tag{2.28}$$

Crank-Nicolson discretization The global semi-discrete system reads $\mathcal{M}\dot{\underline{U}} + \mathcal{S}\underline{U} = \underline{0}$. The θ-method with $\theta = \frac{1}{2}$ is adopted, leading to the Crank-Nicolson scheme:

$$\mathcal{M}\frac{\underline{U}^{k+1} - \underline{U}^k}{\Delta t} + \mathcal{S}\left[\frac{1}{2}\underline{U}^{k+1} + \frac{1}{2}\underline{U}^k\right] = \underline{0} \tag{2.29}$$

Collecting all unknowns on the left-hand side, the linear system to be solved at each time step is:

$$\left(\mathcal{M} + \frac{\Delta t}{2}\mathcal{S}\right)\underline{U}^{k+1} = \left(\mathcal{M} - \frac{\Delta t}{2}\mathcal{S}\right)\underline{U}^{k} \tag{2.30}$$

The scheme is second-order accurate in time and unconditionally stable for the present conservative block system. Since the system matrix on the left-hand side of (2.30) depends only on the mesh and on Δt, it is assembled and factorized once before entering the time loop; only the right-hand side is updated at each step.

Discrete L^2 error The accuracy of the numerical approximation is quantified by the L^2 norm of the error on η at the final time:

$$\|\underline{\eta}_{\text{ex}}(\cdot,T) - \underline{\eta}_h(\cdot,T)\|_{L^2(0,L)} = \left(\int_0^L \left|\underline{\eta}_{\text{ex}}(x,T) - \underline{\eta}_h(x,T)\right|^2 dx\right)^{1/2} \tag{2.31}$$

Within the finite element framework, this integral is evaluated through the global mass matrix $\mathbf{M}$. Letting $\underline{e} = \underline{\eta}_{\text{ex}}(T) - \underline{\eta}_h(T)$ denote the nodal error vector, the discrete estimate of (2.31) is:

$$E_{L^2} = \sqrt{\underline{e}^{\mathrm{T}}\mathbf{M}\underline{e}} \tag{2.32}$$

To ensure that the exact solution is evaluated at the same time actually reached by the solver, the effective time step is reset to $\Delta t_{\text{eff}} = T/N_{\text{steps}}$ with $N_{\text{steps}} = \text{round}(T/\Delta t)$. This correction prevents a mismatch between the target final time and the numerical time from contaminating the measured convergence rates.

2.4.2 Wave Propagation Snapshots

The comparison between the SEM approximation and the exact traveling-wave profile at the final time $T = 0.5\,\text{s}$ is shown in figure 2.1, obtained with $N_e = 20$ elements, polynomial degree $p = 4$ and $\Delta t = 10^{-3}$ s.

Wave packet transport: periodic wrap-around The Gaussian pulse is transported rightward without distortion at the theoretical speed $c = \sqrt{gH}$. The correct enforcement of periodic boundary conditions is evidenced by the re-entry of the pulse from the left boundary after crossing $x = L$. This behavior is reproduced accurately by the SEM solver for both η and q, with no perceptible numerical diffusion.

2.4.3 Convergence Analysis

The spatial convergence of the scheme is assessed by monitoring the discrete error (2.32) under mesh refinement (h-study) and polynomial enrichment (p-study). The theoretical spatial error estimate for the continuous Galerkin SEM of degree p reads:

$$E_{\text{spatial}} \approx Ch^{p+1} \tag{2.33}$$

provided the exact solution is sufficiently smooth. The total measured error also contains the temporal contribution of the Crank-Nicolson scheme:

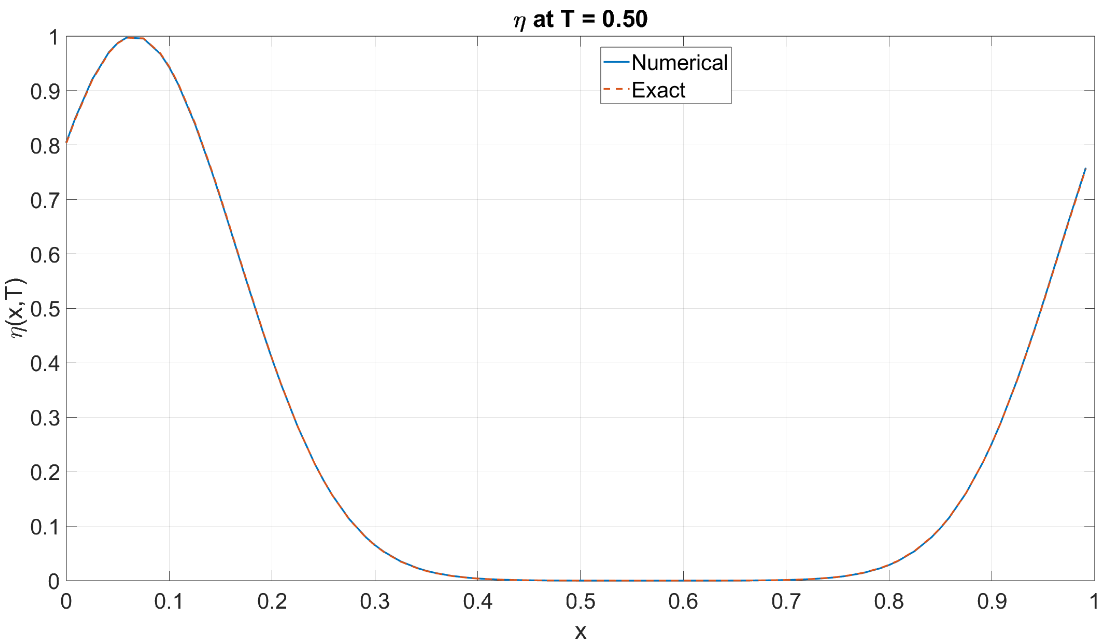
$$E_{\text{total}} \lesssim C_1 h^{p+1} + C_2 \Delta t^2 \tag{2.34}$$

In all convergence tests, Δt is scaled alongside h so that the temporal term remains negligible with respect to the spatial one. The experimental convergence rate between two successive refinements is:

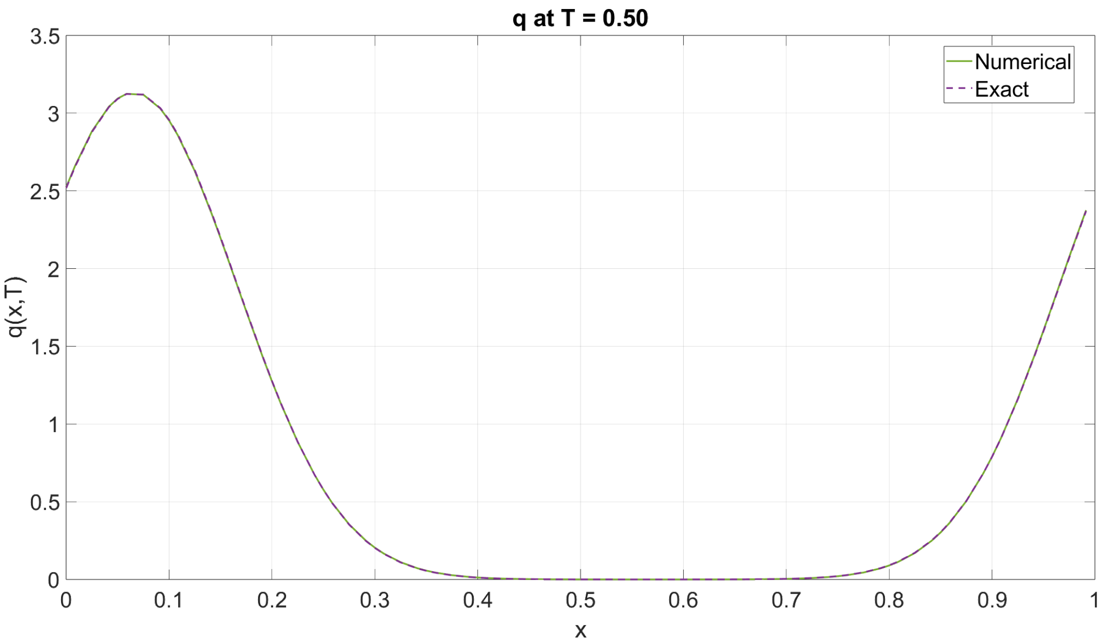
$$r_i = \frac{\log(E_i/E_{i+1})}{\log(h_i/h_{i+1})} \tag{2.35}$$

h-refinement: high-order algebraic convergence The upper panel of figure 2.2 shows that the L^2 error decreases monotonically under mesh refinement, with a slope compatible with the $\mathcal{O}(h^5)$ reference. The slight deviation from the ideal theoretical exponent is physically expected: the Gaussian pulse, when extended periodically through the modulo operation in (2.28), is not globally smooth across the domain boundaries. This limits the regularity assumption underlying estimate (2.33). Furthermore, the second-order temporal error of the Crank-Nicolson scheme contributes an irreducible baseline that prevents the total error from matching the ideal spatial trend exactly.

p-enrichment: monotone decay The lower panel of figure 2.2 confirms that the error decreases monotonically as the polynomial degree p is increased at fixed mesh. The trend is consistent with the high-order accuracy expected from the SEM framework. The same limiting mechanisms identified in the h-study apply here: the limited periodic regularity of the Gaussian datum and the $\mathcal{O}(\Delta t^2)$ temporal floor define the effective accuracy ceiling for any fixed time step. Within these constraints, the observed monotone decay fully validates the spectral element implementation.

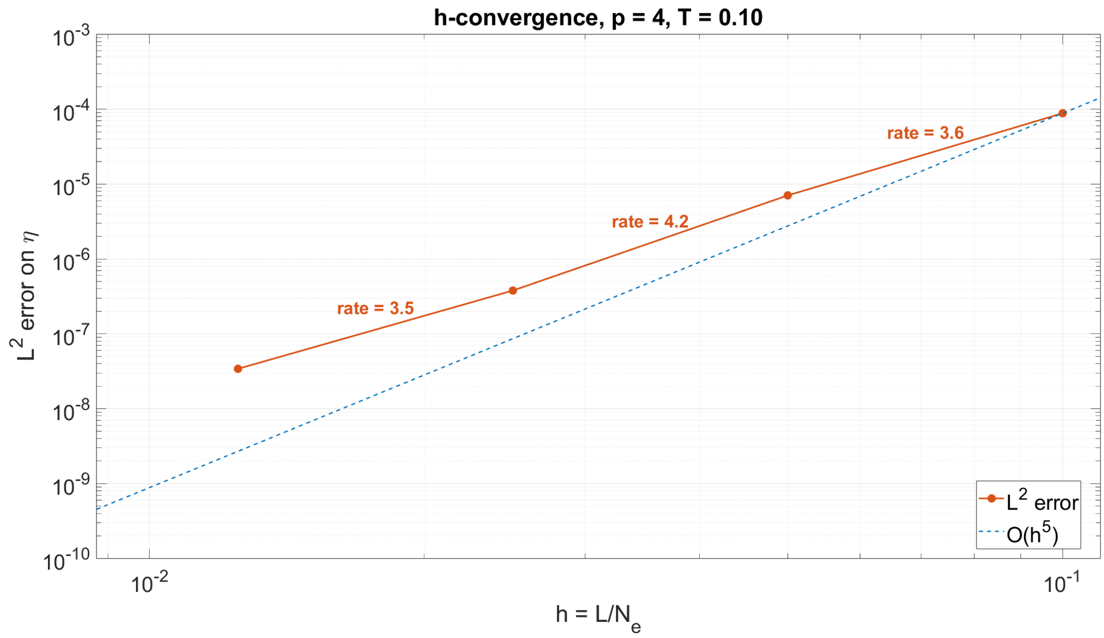


(a) Free-surface elevation $\eta(x, T)$. The numerical profile is visually indistinguishable from the exact solution. Since $cT \approx 1.56\,\mathrm{m} > L$, the pulse has crossed the right boundary and re-entered the domain from the left, providing a direct verification of the periodic connectivity implemented in the solver

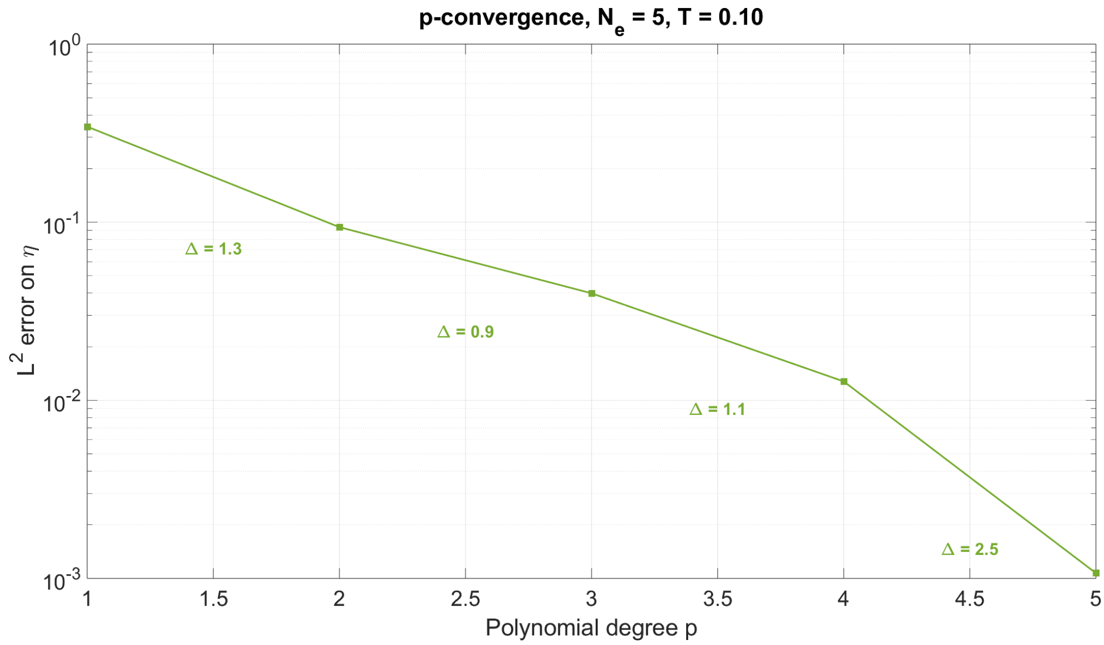


(b) Discharge $q(x, T)$. Since the exact solution satisfies $q_{\text{ex}} = c\eta_{\text{ex}}$ at all times, the discharge reproduces the same translated Gaussian shape. The numerical result captures this proportionality without visible amplitude loss or phase lag, confirming the non-dissipative character of the Crank-Nicolson integrator

Figure 2.1: Wave propagation snapshots for the Gaussian benchmark at final time $T = 0.5\,\mathrm{s}$. Initial conditions as in (2.27), exact solution as in (2.28). Physical parameters: $L = 1\,\mathrm{m}$, $H = 1\,\mathrm{m}$, $g = 9.81\,\mathrm{m/s^2}$. Discretization: $N_e = 20$, $p = 4$, $\Delta t = 10^{-3}$ s. The wrap-around of the pulse from right to left confirms the correct implementation of periodic BC



(a) Log-log plot of E_{L^2} versus mesh size h at fixed polynomial degree $p = 4$, for meshes $N_e = 10, 20, 40, 80$. The error decreases monotonically and follows a high-order algebraic trend. The local experimental rates are slightly below the ideal exponent of 5, consistently with the limited periodic regularity of the Gaussian datum and the residual second-order temporal error of the Crank-Nicolson integrator



(b) Semi-log plot of E_{L^2} versus polynomial degree p at fixed mesh $N_e = 5$, for $p = 1, 2, 3, 4, 5$. The error decays monotonically as the approximation order increases, reflecting the polynomial enrichment of the SEM space. No visible saturation occurs within the tested range

Figure 2.2: Convergence analysis for the Gaussian benchmark. The L^2 error on η is evaluated at $T = 0.5$ s with the same physical parameters as in figure 2.1. The slight deviation from the ideal spectral slopes is physically expected: the Gaussian profile, when extended periodically through the modulo argument, exhibits limited regularity at the domain boundaries, which weakens the smoothness assumptions underlying (2.33). In addition, the second-order temporal error of the Crank-Nicolson scheme provides a non-negligible lower bound on the total error, as expressed in (2.34). Both effects are consistent with the observed behavior

2.5 Studies on Boundary Conditions and Friction

Two optional modifications of the periodic shallow-water model are examined: the replacement of periodicity by reflecting wall boundary conditions and the introduction of a linear friction term in the momentum equation. In both cases, the theoretical implications for the equivalent second-order equation, the energy balance and the discrete block formulation are derived first. The corresponding numerical evidence is then collected and interpreted in a single dedicated subsection.

2.5.1 Part A: Reflecting Wall Boundary Conditions

Strong formulation and equivalent second-order condition The periodic boundary conditions are replaced by rigid-wall conditions, namely

$$q(0,t) = 0, \qquad q(L,t) = 0 \tag{2.36}$$

These impose zero discharge at both endpoints, describing a closed basin where no flux enters or leaves the domain. The equivalent condition for the second-order wave equation is obtained by evaluating the momentum equation $q_t + gH\eta_x = 0$ at $x = 0$ and $x = L$. Since $q(\cdot,t)$ vanishes at both walls for every t, its time derivative also vanishes there. Therefore:

$$q_t(0,t) = 0 \quad \Longrightarrow \quad gH\eta_x(0,t) = 0 \quad \Longrightarrow \quad \eta_x(0,t) = 0 \tag{2.37}$$

and, analogously,

$$q_t(L,t) = 0 \quad \Longrightarrow \quad gH\eta_x(L,t) = 0 \quad \Longrightarrow \quad \eta_x(L,t) = 0 \tag{2.38}$$

Hence, homogeneous Dirichlet conditions on q are equivalent to homogeneous Neumann conditions on η. Physically, the wave is reflected by the wall without changing sign.

Effect on the energy balance The conserved energy introduced in section 2.2 is

$$E(t) = \int_0^L \left(\frac{1}{2}g\eta^2 + \frac{1}{2H}q^2\right) dx \tag{2.39}$$

Repeating the computation carried out for the periodic case yields

$$\frac{dE}{dt} = -g\left[\eta(x,t)q(x,t)\right]_{x=0}^{x=L} = -g\Big(\eta(L,t)q(L,t) - \eta(0,t)q(0,t)\Big) \tag{2.40}$$

Substituting (2.36), the boundary contribution vanishes identically:

$$\frac{dE}{dt} = 0 \tag{2.41}$$

Reflecting walls therefore preserve the total mechanical energy exactly. Unlike the periodic setting, energy is no longer transported across the boundary; instead, it remains trapped inside the domain and is redistributed through successive reflections.

Effect on the weak formulation and discrete spaces The function spaces must be adapted. The periodicity constraint is removed, so the free-surface elevation is sought in

$$V = H^1(0, L) \tag{2.42}$$

Since the discharge must satisfy (2.36) strongly, the corresponding trial and test space becomes

$$V_0 = H_0^1(0, L) = \{v \in H^1(0, L) : v(0) = v(L) = 0\} \tag{2.43}$$

The weak problem reads: find $\eta(\cdot, t) \in V$ and $q(\cdot, t) \in V_0$ such that, for all $v \in V$ and $w \in V_0$,

$$\int_0^L \eta_t v dx + \int_0^L q_x v dx = 0 \tag{2.44}$$

$$\int_0^L q_t w dx + gH \int_0^L \eta_x w dx = 0 \tag{2.45}$$

The algebraic block structure remains the same as in section 2.4; only the treatment of the boundary degrees of freedom changes. In practice, the periodic identification of the first and last node is removed and the endpoint degrees of freedom associated with q are constrained to zero.

2.5.2 Part B: Linear Friction

Modified strong formulation A linear friction term is added to the momentum equation with damping coefficient $\gamma > 0$:

$$\begin{cases} \eta_t + q_x = 0 \\ q_t + gH\eta_x = -\gamma q \end{cases} \qquad \gamma > 0 \tag{2.46}$$

The term $-\gamma q$ models a resistive effect that opposes the discharge and progressively removes mechanical energy from the system.

Equivalent damped wave equation The continuity equation is differentiated with respect to time:

$$\eta_{tt} = -q_{xt} \tag{2.47}$$

The momentum equation is differentiated with respect to x:

$$q_{tx} + gH\eta_{xx} = -\gamma q_x \tag{2.48}$$

By Schwarz's theorem, $q_{xt} = q_{tx}$. Using (2.47) and the identity $q_x = -\eta_t$ from the continuity equation:

$$\eta_{tt} + \gamma\eta_t = gH\eta_{xx} \tag{2.49}$$

This is the Telegrapher's equation. The additional first-order time derivative introduces amplitude attenuation while leaving the wave propagation mechanism otherwise unchanged.

Effect on the energy identity Starting from (2.39), the time derivative of the energy is

$$\frac{dE}{dt} = \int_0^L \left(g\eta\eta_t + \frac{1}{H} q q_t \right) dx \tag{2.50}$$

Substituting the modified system (2.46):

$$\frac{dE}{dt} = \int_0^L \left[g\eta(-q_x) + \frac{1}{H} q(-gH\eta_x - \gamma q) \right] dx \tag{2.51}$$

Collecting terms and recognizing the total derivative $\eta q_x + q\eta_x = \frac{d}{dx}(\eta q)$:

$$\frac{dE}{dt} = -g\left[\eta q\right]_0^L - \frac{\gamma}{H}\int_0^L q^2 dx \tag{2.52}$$

Under either periodic or reflecting-wall boundary conditions, the boundary term vanishes, yielding:

$$\frac{dE}{dt} = -\frac{\gamma}{H}\int_0^L q^2 dx \leq 0 \tag{2.53}$$

Friction therefore destroys the conservative character of the system: the total energy decays monotonically as long as q is not identically zero.

Effect on the discrete block system At the semi-discrete level, the friction term modifies only the $(2,2)$ block of the spatial operator. The conservative block matrix $\mathcal{S}$ is replaced by its damped counterpart:

$$\mathcal{S}_{\text{fr}} = \begin{bmatrix} \mathbf{0} & \mathbf{D} \\ gH\mathbf{D} & \gamma\mathbf{M} \end{bmatrix} \tag{2.54}$$

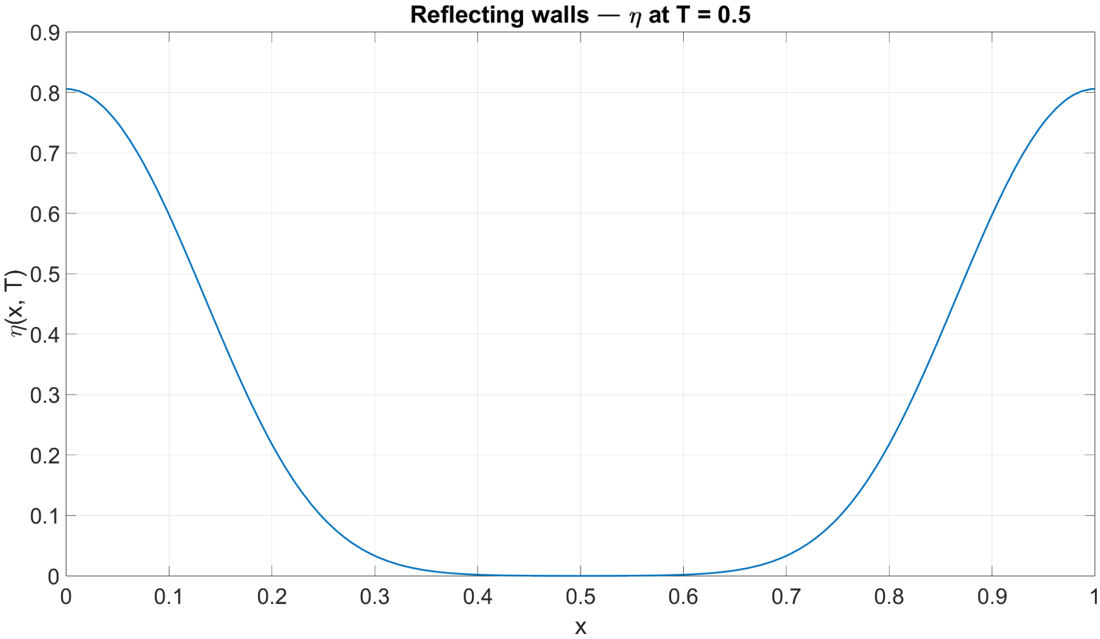
The block mass matrix $\mathcal{M}$ is unchanged. Consequently, the Crank-Nicolson time discretization (2.30) preserves its structure and becomes:

$$\left(\mathcal{M} + \frac{\Delta t}{2}\mathcal{S}_{\text{fr}}\right)\underline{U}^{k+1} = \left(\mathcal{M} - \frac{\Delta t}{2}\mathcal{S}_{\text{fr}}\right)\underline{U}^{k} \tag{2.55}$$

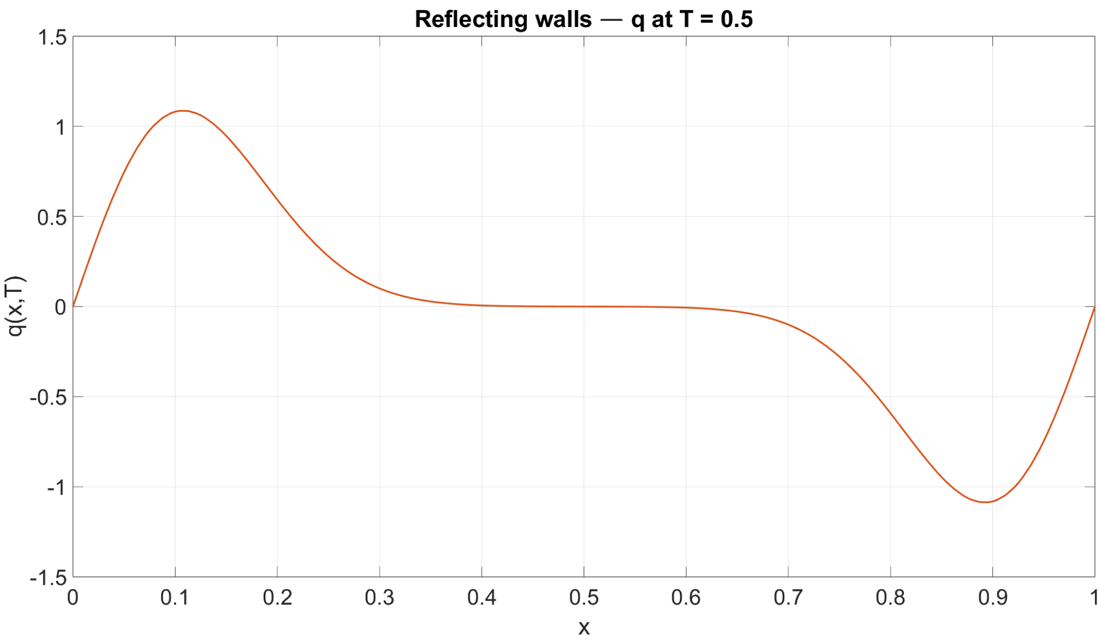
From an implementation standpoint, the friction study requires no new time integrator: it only entails the reassembly of the modified operator (2.54) before entering the time loop.

2.5.3 Numerical Results and Physical Interpretation

Two isolated numerical experiments are conducted using the Gaussian initial data (2.27) on the domain $(0, 1\,\text{m})$ with $H = 1\,\text{m}$, $g = 9.81\,\text{m/s}^2$ and final time $T = 0.5\,\text{s}$. The first experiment imposes reflecting walls with $\gamma = 0$; the second retains periodic boundary conditions and activates linear friction with $\gamma = 1$. In both cases, the finest mesh ($N_e = 80$, $p = 4$) and a sufficiently small time step are employed to avoid temporal error contamination. The Gaussian pulse is the most suitable benchmark for these optional studies: its spatial localization makes both the reflection mechanism and the amplitude decay immediately visible in the computed profiles.



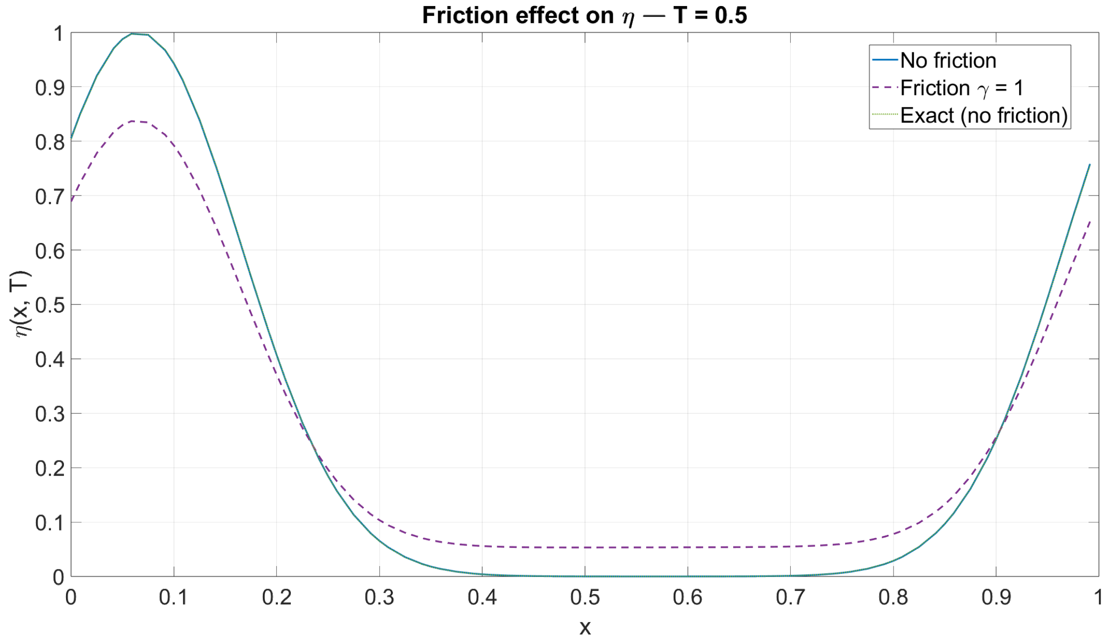
(a) Free-surface elevation $\eta(x, T)$ under reflecting-wall boundary conditions (2.36). The initial rightward-propagating Gaussian pulse has reached the right wall and is reflected back into the domain. The reflected profile retains the original amplitude to a very good approximation, in agreement with the conservative energy identity (2.41)



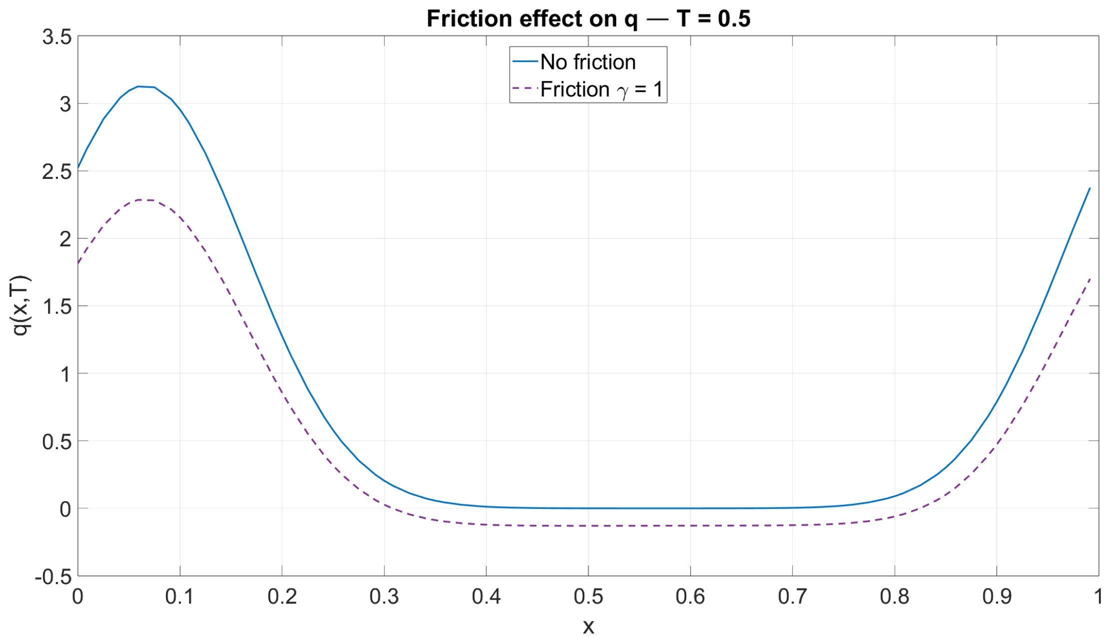
(b) Discharge $q(x, T)$ under reflecting-wall boundary conditions. The discharge goes to exactly zero at both $x = 0$ and $x = L$, enforcing the Dirichlet constraint (2.36). This zero-discharge condition at the walls is the direct mechanism that triggers the conservative in-phase reflection of the free-surface elevation observed in the upper panel

Figure 2.3: Reflecting-wall experiment at $T = 0.5\,\mathrm{s}$. Physical parameters: $L = 1\,\mathrm{m}$, $H = 1\,\mathrm{m}$, $g = 9.81\,\mathrm{m/s^2}$, $\gamma = 0$. Discretization: $N_e = 80$, $p = 4$. The reflected wave preserves its amplitude, confirming that $dE/dt = 0$ in a closed domain.

Reflecting walls: conservative closed-domain dynamics The discharge profile in figure 2.3(b) shows that q is driven to exactly zero at both endpoints, directly enforcing the boundary condition (2.36). This zero-discharge constraint is the physical mechanism responsible for the in-phase reflection of η: the wave cannot exit the domain in terms of flux, so it reverses direction while retaining its amplitude. The numerical profile in figure 2.3(a) confirms the conservation law (2.41): the reflected pulse carries the same peak height as the incident one and no energy is lost at the wall.



(a) Free-surface elevation $\eta(x, T)$ with linear friction $\gamma = 1$ under periodic boundary conditions. The damped solution (solid line) remains in phase with the undamped reference (dashed), confirming that the wave speed $c = \sqrt{gH}$ is unaffected by the friction term. However, the peak amplitude is significantly reduced, visualizing the monotonic energy dissipation expressed by (2.53)



(b) Discharge $q(x, T)$ with linear friction $\gamma = 1$. The discharge profile undergoes the same amplitude attenuation as the elevation, consistently with the proportionality relation $q_{\text{ex}} = c\eta_{\text{ex}}$ that holds in the undamped limit. The reduction in discharge magnitude is directly responsible for the bulk energy loss through the term $-(\gamma/H)\int_0^L q^2\,dx$

Figure 2.4: Friction experiment at $T = 0.5\,\text{s}$ with linear damping coefficient $\gamma = 1$. Physical parameters and discretization as in figure 2.3. The damped wave remains in phase with the undamped reference while exhibiting a clear amplitude reduction, in agreement with the Telegrapher's equation (2.49) and the energy inequality (2.53)

Friction: dissipative transport Figure 2.4 shows that the linear friction term does not alter the propagation speed: the damped profile remains perfectly in phase with the undamped reference. The effect of friction is exclusively dissipative, manifesting as a progressive reduction of the wave amplitude. Both η and q are attenuated by the same factor, consistently with the energy law (2.53). The Telegrapher's equation (2.49) provides the analytical explanation: the additional first-order time derivative $\gamma\eta_t$ introduces exponential amplitude decay while leaving the wave speed c unchanged.

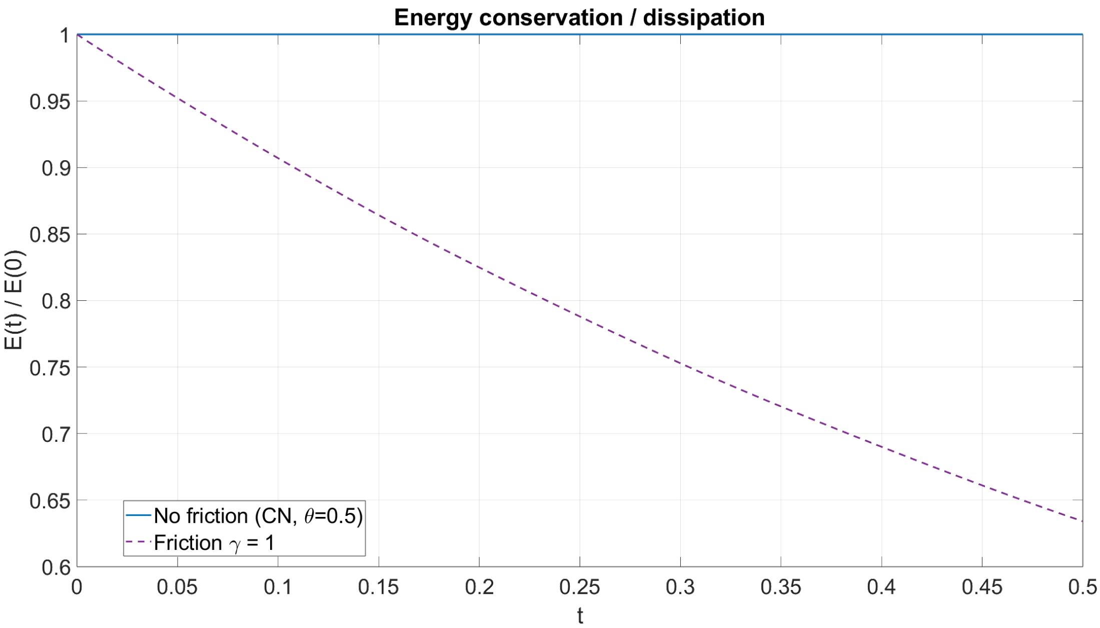


Figure 2.5: Time evolution of the total discrete energy $E(t) = \sum_i \left(\frac{1}{2}g\eta_i^2 + \frac{1}{2H}q_i^2\right)\Delta x_i$ for the Gaussian benchmark under two configurations: without friction ($\gamma = 0$, solid line) and with linear friction ($\gamma = 1$, dashed line). Physical parameters and discretization as in figure 2.3. The flat energy curve for $\gamma = 0$ confirms that the combination of the SEM spatial discretization and the Crank-Nicolson time integrator preserves the discrete energy identity (2.41) to machine precision. The monotonically decreasing curve for $\gamma = 1$ is consistent with the dissipation law (2.53), providing quantitative numerical validation of the analytical derivation

Energy analysis: validation of the analytical predictions Figure 2.5 provides the conclusive numerical validation of the analytical derivations carried out in sections 2.5.1 and 2.5.2. In the frictionless case ($\gamma = 0$), the total discrete energy remains constant throughout the simulation: the flat line confirms that the Crank-Nicolson scheme, combined with the SEM spatial discretization, preserves the conservative energy identity (2.41) to machine precision. This is a non-trivial property: it reflects the symplectic-like character of the Crank-Nicolson integrator when applied to a skew-symmetric spatial operator. When friction is activated ($\gamma = 1$), the energy curve decreases monotonically from its initial value, in full agreement with the theoretical inequality (2.53). The rate of decay is governed by the bulk dissipation term $-(\gamma/H)\int_0^L q^2 dx$, which is largest when the discharge is at its maximum and tapers off as the wave amplitude diminishes over time.

3

Homework 3

3.1 Finite Volume Approximation

The third homework addresses the numerical solution of the one-dimensional nonlinear Shallow Water Equations (SWE) by means of the Finite Volume Method (FVM), comparing a first-order Godunov scheme and a second-order Lax-Wendroff reconstruction.

3.1.1 The Mathematical Model

The propagation of water waves in the shallow-water regime is governed, in its full nonlinear formulation, by a system of hyperbolic conservation laws that couples the evolution of the water height $h(x,t)$ and the depth-averaged discharge $q(x,t) = h(x,t)u(x,t)$, where $u(x,t)$ denotes the depth-averaged horizontal velocity. Unlike the linearized shallow water equations treated in the previous chapter, no small-amplitude assumption is invoked here, and the full nonlinear dynamics must be retained. The governing system on the spatial domain $(0, L)$ over the time interval $(0, T]$, with gravitational acceleration g, reads

$$\begin{cases} h_t + (hu)_x = 0 \\ (hu)_t + \left(hu^2 + \dfrac{1}{2}gh^2\right)_x = 0 \end{cases} \tag{3.1}$$

where the first equation expresses conservation of mass and the second expresses conservation of momentum. Introducing the state vector $\underline{U} = [h, q]^T$ and the flux vector $\underline{F}(\underline{U})$, system (3.1) admits the compact conservative form

$$\underline{U}_t + \underline{F}(\underline{U})_x = \underline{0} \tag{3.2}$$

where the two vectors are explicitly defined as

$$\underline{U} = \begin{bmatrix} h \\ q \end{bmatrix}, \qquad \underline{F}(\underline{U}) = \begin{bmatrix} q \\ \dfrac{q^2}{h} + \dfrac{1}{2}gh^2 \end{bmatrix} \tag{3.3}$$

The nonlinearity of $\underline{F}$ has a fundamental consequence on the solution structure. The propagation speed of any perturbation is no longer a fixed quantity but depends on the local values of h and u: the two characteristic speeds of system (3.2) are $\lambda_{1,2} = u \mp c$, where $c = \sqrt{gh}$ is the local wave celerity. Because these speeds vary with the solution itself, characteristic curves emanating from different points can intersect in finite time, even when the initial data are smooth. At the crossing point, classical pointwise derivatives cease to exist and a shock, a physical discontinuity in both h and q, forms spontaneously. Classical strong-form numerical methods based on nodal derivatives (finite differences, finite elements, spectral elements) are therefore inadequate in the vicinity of shocks: they either generate spurious non-physical oscillations or converge to weak solutions that violate the entropy condition and are thus physically incorrect. The conservative form (3.2) provides the correct mathematical framework, since it remains meaningful in the distributional sense across discontinuities and directly encodes the integral conservation laws from which the physical jump conditions (Rankine-Hugoniot relations) are derived.

3.1.2 Characteristic Structure and Hyperbolicity

The design of any upwind numerical flux requires knowledge of the characteristic speeds of the system (3.2). These are the eigenvalues of the flux Jacobian $\mathbf{A}(U) = \partial F/\partial U$. Differentiating the flux vector $F(U) = [q, q^2/h + \frac{1}{2}gh^2]^T$ with respect to $U = [h, q]^T$, and expressing the result in terms of the depth-averaged velocity $u = q/h$, one obtains

$$\mathbf{A}(U) = \frac{\partial F}{\partial U} = \begin{pmatrix} \dfrac{\partial q}{\partial h} & \dfrac{\partial q}{\partial q} \\ \dfrac{\partial}{\partial h}\left(\dfrac{q^2}{h} + \dfrac{1}{2}gh^2\right) & \dfrac{\partial}{\partial q}\left(\dfrac{q^2}{h} + \dfrac{1}{2}gh^2\right) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ gh - u^2 & 2u \end{pmatrix} \tag{3.4}$$

The eigenvalues of $\mathbf{A}(U)$ are found from $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$:

$$\lambda^2 - 2u\lambda - (gh - u^2) = 0 \quad \Longrightarrow \quad \lambda_{1,2} = u \mp c, \qquad c = \sqrt{gh} \tag{3.5}$$

where $c = \sqrt{gh}$ is the local shallow-water wave celerity. Since $\lambda_1 \neq \lambda_2$ for all $h > 0$, the system is strictly hyperbolic in the wet domain. The two eigenvalues $\lambda_1 = u - c$ and $\lambda_2 = u + c$ represent the speeds of left- and right-propagating gravity waves respectively, superimposed on the advective velocity u.

Physical interpretation of the characteristic speeds. The sign structure of $\lambda_{1,2}$ classifies the local flow regime:

- Subcritical flow ($|u| < c$, Froude number $\mathrm{Fr} = |u|/c < 1$): $\lambda_1 < 0 < \lambda_2$, so one wave travels leftward and one rightward. Information propagates in both directions.
- Supercritical flow ($|u| > c$, $\mathrm{Fr} > 1$): both eigenvalues have the same sign, all information propagates in a single direction.
- Critical flow ($|u| = c$, $\mathrm{Fr} = 1$): one eigenvalue vanishes, signaling the occurrence of a sonic point. Special care is required at sonic points, where linearized Riemann solvers may converge to an entropy-violating solution; the exact Godunov flux adopted here samples the transonic rarefaction fan directly and therefore selects the correct entropy-satisfying state without any additional fix.

Because the characteristic speeds $\lambda_{1,2}$ depend on h and u, which are themselves unknowns evolving in time, characteristics emanating from distinct spatial points can intersect in finite time. At the crossing point, the classical solution breaks down and a shock forms spontaneously, even from smooth initial data. This motivates the use of the conservative FVM framework, which remains well-defined across discontinuities.

3.1.3 FVM Discretization

The Finite Volume Method is the natural discretization framework for conservation laws in the form (3.2), as it operates directly on integral cell averages rather than pointwise values, thereby preserving conservation at the discrete level and remaining well-defined across discontinuities. The spatial domain $[0, L]$ is partitioned into M uniform cells of width $\Delta x =$

L/M. The i-th cell is the interval $[x_{i-1/2}, x_{i+1/2}]$, with center $x_i = (i - \frac{1}{2})\Delta x$ for $i = 1, \dots, M$. The time step Δt is recomputed at every iteration from the CFL stability condition. For explicit finite-volume schemes applied to hyperbolic systems, the CFL condition requires that no wave travels more than one cell width per time step:

$$\Delta t^n = \nu \frac{\Delta x}{\max_i S_i^n}, \qquad S_i^n = |u_i^n| + c_i^n = \left|\frac{q_i^n}{h_i^n}\right| + \sqrt{g h_i^n} \tag{3.6}$$

where $\nu \in (0, 1)$ is the Courant number, set to $\nu = 0.5$ in all simulations. The maximum wave speed $\max_i S_i^n$ must be evaluated at each time level t^n from the current numerical solution, since h and u evolve nonlinearly and the local wave celerities $c_i = \sqrt{g h_i}$ change accordingly. A fixed time step would either violate stability when fast waves develop near the shock, or be unnecessarily conservative throughout the simulation. The adaptive formula (3.6) ensures that the CFL constraint is satisfied at every time level with the largest permissible Δt^n, minimizing the total number of time steps while preserving stability.The fundamental discrete unknown is the cell average of the state vector at time t^n:

$$\underline{U}_i^n \approx \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} \underline{U}(x, t^n) dx \tag{3.7}$$

Integrating the conservation law (3.2) over the cell $[x_{i-1/2}, x_{i+1/2}]$ and applying the divergence theorem yields, after division by Δx, the exact integral balance

$$\frac{d}{dt}\underline{U}_i(t) = -\frac{1}{\Delta x}\left[\underline{F}\left(\underline{U}(x_{i+1/2}, t)\right) - \underline{F}\left(\underline{U}(x_{i-1/2}, t)\right)\right] \tag{3.8}$$

Replacing the continuous interface fluxes with numerical approximations $\underline{F}_{i+1/2}^n$ and applying a forward-Euler time step to (3.8) gives the fully discrete conservative update

$$\underline{U}_i^{n+1} = \underline{U}_i^n - \frac{\Delta t}{\Delta x}\left[\underline{F}_{i+1/2}^n - \underline{F}_{i-1/2}^n\right] \tag{3.9}$$

Equation (3.9) expresses the key property of the FVM: the change in the cell average $\underline{U}_i$ between two time levels depends exclusively on the numerical fluxes at the two bounding interfaces $x_{i\pm1/2}$. Since every interface flux entering cell i also leaves cell $i+1$ with the opposite sign, the total conserved quantities $\sum_i h_i$ and $\sum_i q_i$ are preserved to machine precision, irrespective of the spatial or temporal resolution. Reflecting wall boundary conditions are enforced at both endpoints through a ghost-cell approach: the ghost cells mirror the water height of the adjacent interior cell while negating the discharge, so that $q = 0$ is imposed at the domain boundaries consistently with the physical requirement of an impenetrable wall.

3.1.4 First-Order Godunov Scheme

The Godunov scheme achieves first-order accuracy in both space and time by adopting the simplest consistent reconstruction of the solution within each cell: the state is assumed to be piecewise constant, equal to the cell average $\underline{U}_i^n$ throughout the entire cell $[x_{i-1/2}, x_{i+1/2}]$ at each time level t^n. Under this assumption, the solution profile at any interface $x_{i+1/2}$ exhibits an instantaneous jump from the left state $\underline{U}_i^n$ to the right state $\underline{U}_{i+1}^n$. This configuration constitutes a classical Riemann problem, whose exact solution describes the self-similar wave pattern consisting of shocks, contact discontinuities, and rarefaction fans that develops instantaneously from the initial discontinuity. The Godunov numerical flux is defined as the exact physical flux evaluated at the solution of this local Riemann problem:

$$\underline{F}^{\text{God}}_{i+1/2} = \underline{F}\left(\underline{U}^*\left(\underline{U}^n_i, \underline{U}^n_{i+1}\right)\right) \tag{3.10}$$

The exact solution of the Riemann problem for the nonlinear SWE has a two-wave structure: a left and a right wave, each of which is either a shock or a rarefaction, separated by a constant intermediate state $\underline{U}^* = [h^*, h^*u^*]^T$. The intermediate depth h^* is the root of the algebraic equation

$$f_L(h^*, h_L) + f_R(h^*, h_R) + (u_R - u_L) = 0 \tag{3.11}$$

$$f_k(h, h_k) = \begin{cases} 2\left(\sqrt{gh} - \sqrt{gh_k}\right) & h \le h_k \quad \text{(rarefaction)} \\ (h - h_k)\sqrt{\dfrac{g}{2}\left(\dfrac{1}{h} + \dfrac{1}{h_k}\right)} & h > h_k \quad \text{(shock)} \end{cases} \tag{3.12}$$

which is solved numerically for h^*, after which the intermediate velocity follows as $u^* = \frac{1}{2}(u_L + u_R) + \frac{1}{2}\left[f_R(h^*, h_R) - f_L(h^*, h_L)\right]$. Once $\underline{U}^*$ is known, the self-similar solution $\underline{U}(x/t)$ is sampled at the cell interface $\xi = x/t = 0$, selecting the state that occupies the interface according to the speeds of the two waves, and the Godunov flux (3.10) is evaluated on that sampled state. Because the Riemann problem is solved exactly, the scheme transports information along the correct characteristic directions and automatically selects the entropy-satisfying solution, including the transonic case in which a rarefaction fan straddles the interface. The first-order Godunov scheme is therefore monotone and non-oscillatory across discontinuities, capturing both the shock speed and the post-shock state correctly; its only drawback is the numerical diffusion intrinsic to the piecewise-constant reconstruction, which smears shocks and rarefactions over $\mathcal{O}(\Delta x)$ cells.

3.1.5 Second-Order Lax-Wendroff Reconstruction

The dissipation inherent to the Godunov scheme can be substantially reduced by improving the spatial accuracy of the reconstruction from first to second order. Instead of assuming a constant state within each cell, a piecewise linear reconstruction is adopted: the solution inside each cell is represented as a linear polynomial whose slope is inferred from the difference between neighboring cell averages. The reconstructed values at the left and right sides of the interface $x_{i+1/2}$, denoted $\underline{U}^-_{i+1/2}$ (approaching from cell i) and $\underline{U}^+_{i+1/2}$ (approaching from cell $i+1$), are obtained by extrapolating the respective cell averages to the interface:

$$\underline{U}^-_{i+1/2} = \underline{U}_i + \frac{1}{2}\left(\underline{U}_{i+1} - \underline{U}_i\right) \tag{3.13}$$

$$\underline{U}^+_{i+1/2} = \underline{U}_{i+1} - \frac{1}{2}\left(\underline{U}_{i+1} - \underline{U}_i\right) \tag{3.14}$$

In the limit $\underline{U}_{i+1} \to \underline{U}_i$, both reconstructed states reduce to the common cell value, recovering the Godunov piecewise-constant reconstruction as expected. For a general jump, the two reconstructed states are passed to the Riemann solver in place of the raw cell averages, yielding the second-order numerical flux

$$\underline{F}^{\text{LW}}_{i+1/2} = \underline{F}^*\left(\underline{U}^-_{i+1/2}, \underline{U}^+_{i+1/2}\right) \tag{3.15}$$

where $\underline{F}^*$ denotes the numerical flux returned by the Riemann solver evaluated at the reconstructed interface states. The piecewise-linear reconstruction reduces the truncation error from $\mathcal{O}(\Delta x)$ to $\mathcal{O}(\Delta x^2)$ in smooth regions, yielding significantly sharper resolution of both smooth waves and post-shock profiles. However, the higher-order reconstruction introduces a dispersive error component: near sharp discontinuities, where the linear slope approximation is not bounded, the method may generate spurious oscillations (Gibbs-like ripples) that propagate away from the shock front. This trade-off between accuracy in smooth regions and spurious oscillations at discontinuities motivates the introduction of slope limiters such as the minmod or van Leer limiters, which are however beyond the scope of the present work.

3.2 Riemann Problems and Physical Interpretation

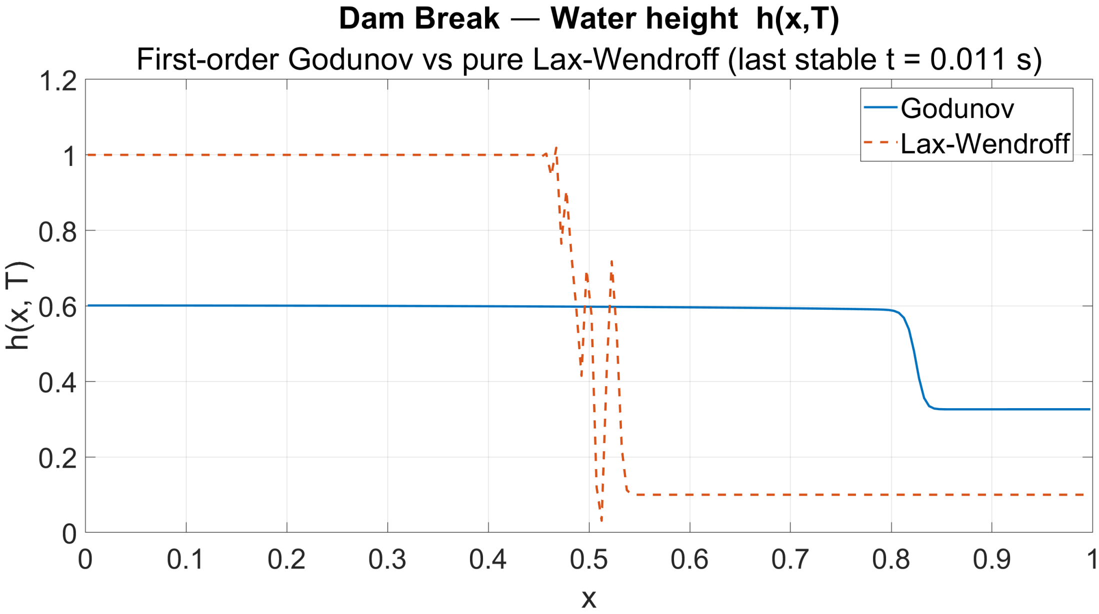
All three Riemann problems are solved on the domain $(0, L)$ with $L = 1\,\mathrm{m}$, $g = 9.81\,\mathrm{m/s^2}$ and final time $T = 1\,\mathrm{s}$, employing a uniform FVM mesh. In all three test cases the discharge q is driven to zero at both $x = 0$ and $x = L$ at every time level, confirming the correct implementation of the reflecting wall boundary conditions enforced through the ghost-cell approach described in section 3.1.3.

3.2.1 Dam Break and Reverse Dam Break

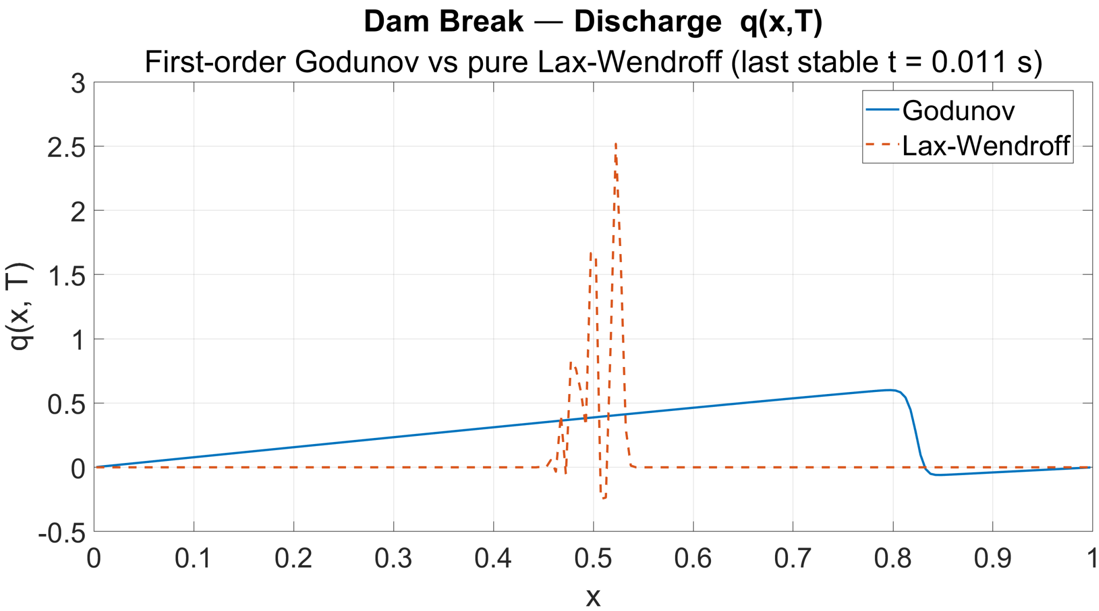
Dam break: physical structure The dam-break problem is initialized with water height $h_L = 1$ on the left half of the domain and $h_R = 0.1$ on the right. The initial discontinuity at $x = L/2$ immediately generates two wave families that propagate in opposite directions. A smooth rarefaction fan travels leftward, continuously expanding the region of intermediate height and gradually accelerating the fluid. A shock wave travels rightward as a steep, near-vertical front, carrying the abrupt transition between the intermediate state and the undisturbed low water on the right. The discharge q is positive throughout the domain, reflecting the net mass transport from the high reservoir towards the low region. After reaching the right wall, the shock reflects and the discharge drops back to zero at the boundary. These features are all visible in the Godunov solution of figure 3.1.

Dam break: numerical comparison The Godunov scheme captures the shock in a rigorously monotone fashion: no overshoot or undershoot appears, and the solution satisfies the entropy condition by construction. The price paid is significant numerical diffusion, the shock front, which is a true mathematical discontinuity, is spread over several mesh cells, and the rarefaction fan is similarly smeared. The pure Lax-Wendroff reconstruction behaves very differently. Lacking any slope limiter, the linear reconstruction is unbounded at the initial discontinuity: spurious oscillations appear immediately at $x = L/2$ and grow without control, driving the reconstructed height out of the physically admissible range. The computation becomes unstable already at $t \approx 0.011\,\mathrm{s}$, long before the final time, and figure 3.1 reports the solution at this last stable instant. This blow-up is not a defect of the implementation but the expected behavior of an unlimited second-order scheme on discontinuous data, and it is precisely what motivates the slope limiters discussed in section 3.3.3.

Reverse dam break: mirror dynamics The reverse dam-break configuration, initialized with $h_L = 0.1$ and $h_R = 1$, produces a wave structure that is the spatial mirror image of the standard dam break. The shock now travels leftward from the initial discontinuity, while the rarefaction fan propagates to the right. The discharge q is uniformly negative, indicating mass transport from right to left, and reaches zero at both walls as required by the boundary conditions. The Godunov scheme reproduces this mirrored structure monotonically, while the pure Lax-Wendroff scheme again becomes unstable at the initial discontinuity and breaks



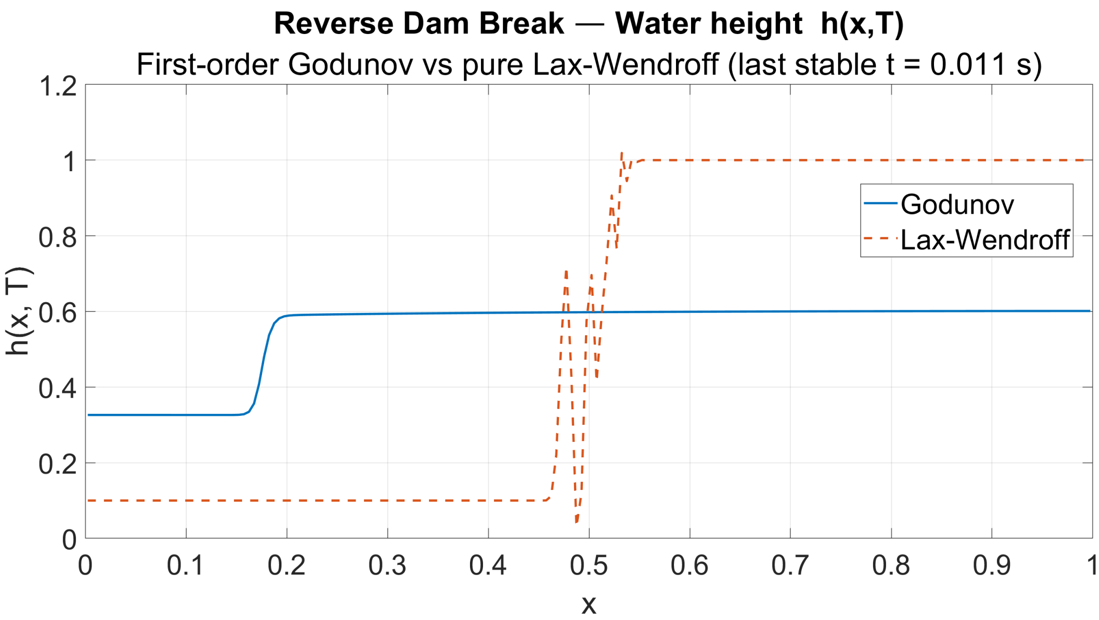
(a) Water height $h(x, T)$. The Godunov scheme (solid blue) resolves the leftward rarefaction and the rightward shock monotonically, with fronts diffused over a few cells. The pure Lax-Wendroff reconstruction (dashed orange) is shown at its last stable time $t \approx 0.011$ s: without a slope limiter it develops violent oscillations at the initial discontinuity and the height locally leaves the physical range, after which the computation breaks down well before $T = 1$ s



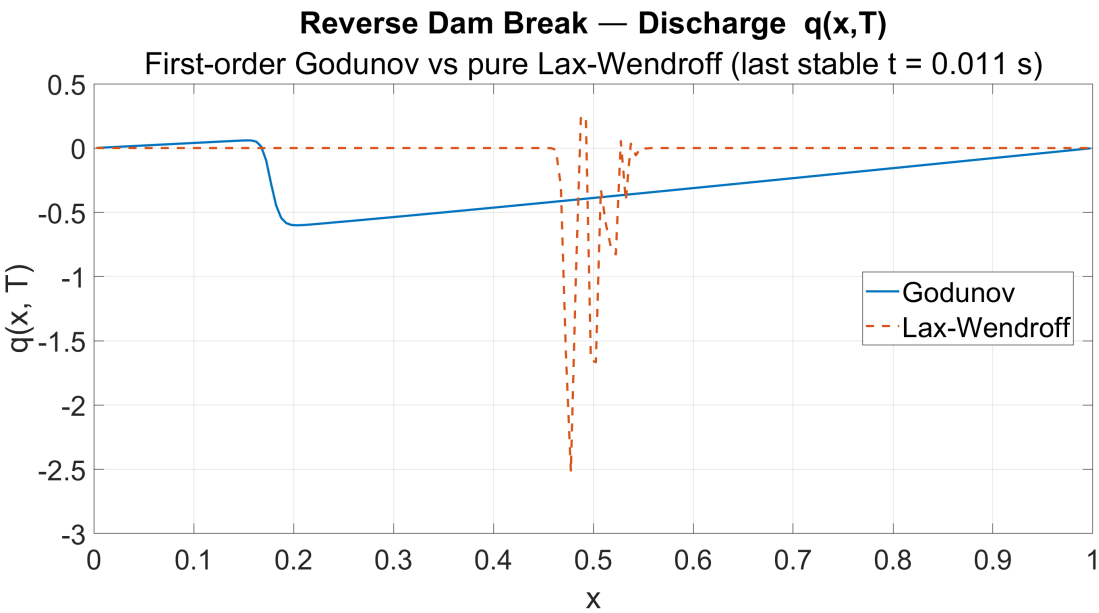
(b) Discharge $q(x, T)$. The Godunov solution is positive throughout, confirming the net rightward mass transport from the high reservoir to the low region, and returns to zero at both walls in agreement with the reflecting boundary conditions. The Lax-Wendroff curve again diverges near the center, with spurious spikes of q that have no physical counterpart

Figure 3.1: Dam-break Riemann problem at $T = 1$ s: first-order Godunov versus pure (unlimited) Lax-Wendroff. The pure Lax-Wendroff scheme is unstable on discontinuous data and is plotted at its last stable time; the comparison motivates the slope limiters discussed in section 3.3.3. Physical parameters: $L = 1$ m, $g = 9.81$ m/s^2, $h_L = 1$, $h_R = 0.1$

down at $t \approx 0.011$ s, confirming that the numerical behavior of the two methods is symmetric with respect to the direction of propagation.



(a) Water height $h(x, T)$. The shock now propagates leftward, mirroring the standard dam break; the Godunov scheme (solid blue) captures it monotonically. The pure Lax-Wendroff reconstruction (dashed orange), shown at its last stable time $t \approx 0.011$ s, again develops uncontrolled oscillations at the initial discontinuity before breaking down



(b) Discharge $q(x, T)$. The Godunov values are negative throughout, confirming the leftward mass transport, with $q = 0$ enforced at both walls. The Lax-Wendroff solution diverges near the center, with the same spurious spikes observed in the forward case

Figure 3.2: Reverse dam-break Riemann problem at $T = 1$ s: first-order Godunov versus pure (unlimited) Lax-Wendroff. The structure is the spatial mirror image of the standard dam break, and the pure Lax-Wendroff scheme is again plotted at its last stable time. Physical parameters: $L = 1\,\mathrm{m}$, $g = 9.81\,\mathrm{m/s^2}$, $h_L = 0.1$, $h_R = 1$

3.2.2 Stationary Shock (Hydraulic Jump)

Rankine-Hugoniot analysis for case (c) Before discussing the numerical results, it is essential to examine whether the initial data prescribed in case (c), namely

$$h(x,0) = \begin{cases} 1 & x < \frac{1}{2} \\ 0.5 & x \geq \frac{1}{2} \end{cases}, \qquad u(x,0) = 0 \quad \Longrightarrow \quad q(x,0) = h(x,0)u(x,0) = 0 \tag{3.16}$$

are consistent with a stationary shock. A shock of speed σ separating left state $\underline{U}_L = [h_L, q_L]^T$ from right state $\underline{U}_R = [h_R, q_R]^T$ must satisfy the Rankine-Hugoniot jump conditions:

$$\sigma[h] = [q], \qquad \sigma[q] = \left[\frac{q^2}{h} + \frac{1}{2}gh^2\right] \tag{3.17}$$

where $[\cdot] = (\cdot)_R - (\cdot)_L$ denotes the jump across the discontinuity. Substituting $q_L = q_R = 0$ into the first equation of (3.17):

$$\sigma(h_R - h_L) = 0 \tag{3.18}$$

Since $h_L = 1 \neq h_R = 0.5$, this is satisfied only if $\sigma = 0$. Substituting $\sigma = 0$ and $q_L = q_R = 0$ into the second equation of (3.17):

$$0 = \frac{1}{2}g\left(h_R^2 - h_L^2\right) = \frac{1}{2} \times 9.81 \times (0.25 - 1) = -3.68\,\mathrm{m}^2/\mathrm{s}^2 \neq 0 \tag{3.19}$$

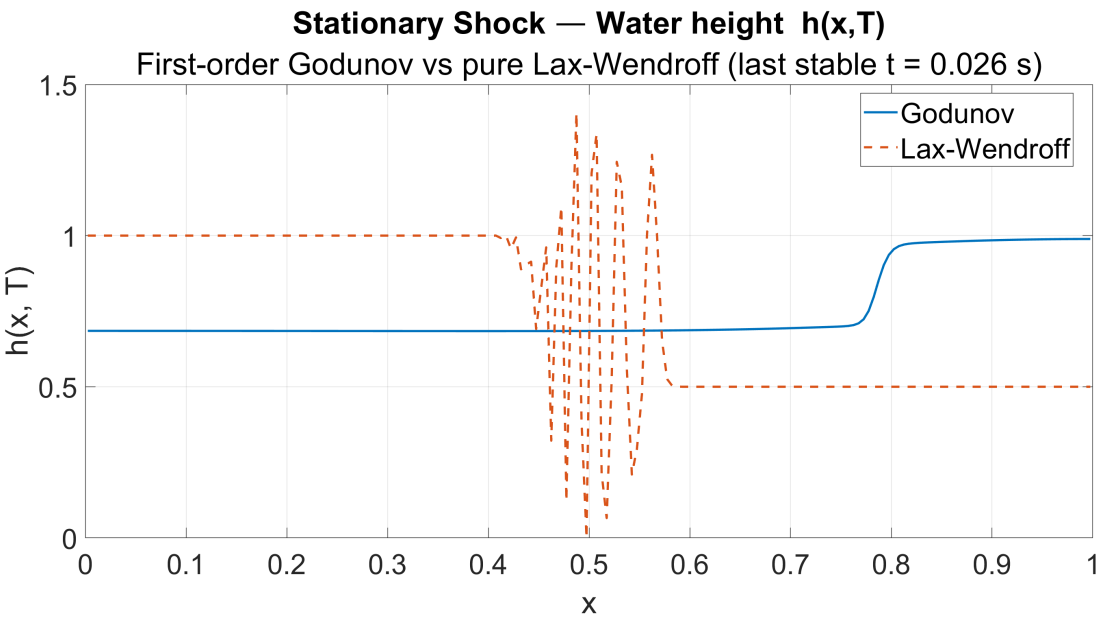
Equation (3.19) shows that the second Rankine-Hugoniot condition is not satisfied for $\sigma = 0$ with $q_L = q_R = 0$. The initial data (3.16) therefore do not constitute a stationary shock: the pressure imbalance $\frac{1}{2}g(h_L^2 - h_R^2) \neq 0$ immediately drives a net momentum flux across the interface. The resulting Riemann problem generates two outgoing wave families, a left-propagating rarefaction fan and a right-propagating shock, which subsequently interact with the reflecting walls. What appears near the center of the domain at $T = 1$ s is not a true stationary hydraulic jump but rather the complex wave pattern arising from the shock reflecting off the right wall and re-entering the domain. A true stationary hydraulic jump would require a non-zero incoming flow satisfying

$$q_L = q_R \neq 0, \qquad \frac{q_L^2}{h_L} + \frac{1}{2}gh_L^2 = \frac{q_R^2}{h_R} + \frac{1}{2}gh_R^2 \tag{3.20}$$

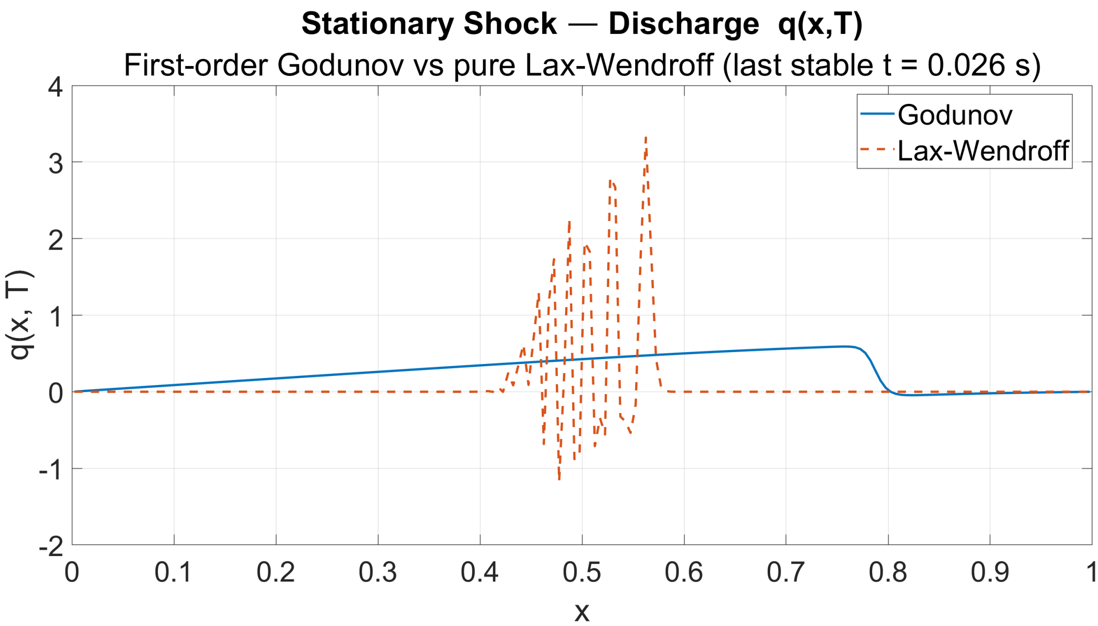
which is not the case here.

Physical structure of the transient Consistently with the analysis above, the solution at $T = 1$ s is not a force-balanced stationary jump but the late stage of the transient generated by the unbalanced initial data. The Godunov solution in figure 3.3 shows a sharp transition in water height that has migrated away from the center and now sits near $x \approx 0.78$, the result of the right-going shock having reflected off the wall and re-entered the domain. The discharge is not identically zero in the interior: it rises from the left wall, reaches a maximum, and drops across the transition before returning to zero at the right wall, a profile characteristic of an actively evolving flow rather than a steady state. Locally, the transition still separates a faster, shallower region on its left from a slower, deeper region on its right, so the supercritical-to-subcritical reading of a hydraulic jump remains a useful description of the front itself, even though the global configuration is not stationary.

The most demanding test for the unlimited scheme The stationary-shock case is the hardest of the three for the pure Lax-Wendroff reconstruction. Because the strongest gradient is anchored near the center of the domain rather than advecting across the mesh, the unbounded linear reconstruction is excited there at every step and the oscillations amplify coherently, reaching the largest amplitude of the three tests. As reported in figure 3.3, the



(a) Water height $h(x, T)$. The Godunov scheme (solid blue) shows a sharp transition that has migrated to $x \approx 0.78$, confirming that the front is not stationary. The pure Lax-Wendroff reconstruction (dashed orange) is shown at its last stable time $t \approx 0.026$ s; this case is the most demanding for the unlimited scheme, and the oscillations at the initial discontinuity reach the largest amplitude observed in the three tests before the computation breaks down



(b) Discharge $q(x, T)$. The Godunov profile rises from the left wall, peaks, and drops across the transition before returning to zero at the right wall, a signature of an actively evolving flow rather than a steady state. The Lax-Wendroff solution diverges near the center

Figure 3.3: Stationary-shock Riemann problem (case c) at $T = 1$ s: first-order Godunov versus pure (unlimited) Lax-Wendroff. Despite the name of the test case, the zero-momentum initial data do not produce a stationary jump (see the Rankine-Hugoniot analysis above); the Godunov front has reflected off the right wall and re-entered the domain. The pure Lax-Wendroff scheme is plotted at its last stable time. Physical parameters: $L = 1$ m, $g = 9.81$ m/s^2, $h_L = 1$, $h_R = 0.5$, reflecting walls at both ends

scheme loses stability already at $t \approx 0.026$ s. The Godunov scheme, by contrast, remains entirely monotone throughout the simulation and tracks the reflected front without spurious oscillations, confirming the robustness of the exact Riemann solver discussed in section

3.1.4. Taken together with the two dam-break tests, this comparison shows that an unlimited second-order reconstruction is unusable on problems containing discontinuities; the remedy is discussed in section 3.3.3.

Quantitative wave-speed estimates for the dam break The exact Riemann solution provides a quantitative check on the dam-break results. The initial celerities on each side of the discontinuity are

$$c_L = \sqrt{gh_L} = \sqrt{9.81 \times 1} \approx 3.13\,\mathrm{m/s}, \qquad c_R = \sqrt{gh_R} = \sqrt{9.81 \times 0.1} \approx 0.99\,\mathrm{m/s} \tag{3.21}$$

Solving the Riemann problem for the dam-break data $(h_L, h_R) = (1, 0.1)$ with $q_L = q_R = 0$ gives the intermediate state $h^* \approx 0.39$, $u^* \approx 2.32\,\mathrm{m/s}$, and a right-going shock of speed $\sigma \approx 3.1\,\mathrm{m/s}$. The shock therefore reaches the right wall (distance $L/2 = 0.5\,\mathrm{m}$) at $t_R \approx 0.5/3.1 \approx 0.16\,\mathrm{s}$, well before $T = 1\,\mathrm{s}$; this is consistent with figure 3.1, where at the final time the Godunov shock has already reflected off the wall and the discharge has returned to zero at the boundary. The left-going rarefaction fan spreads at speeds up to $c_L \approx 3.13\,\mathrm{m/s}$, reaching the left wall on a comparable timescale. For the reverse dam break the same analysis applies by symmetry under $h_L \leftrightarrow h_R$: the shock propagates leftward at the same speed, and the figures confirm the mirror behavior.

3.3 Modified Pressure Flux

The standard SWE system (3.2) assumes a hydrostatic pressure law of the form $p(h) = \frac{1}{2}gh^2$, which leads to the quadratic pressure term in the second component of the flux vector (3.3). This section considers a modification in which the pressure law is replaced by a logarithmic form, giving rise to an alternative conservative system whose mathematical structure and wave dynamics differ in a physically meaningful way from the standard model.

3.3.1 Analytical Formulation

The modified pressure law is defined as

$$p(h) = gh \log\left(\frac{h}{h_{\mathrm{ref}}}\right) \tag{3.22}$$

where $h_{\mathrm{ref}} > 0$ is a fixed reference height. Substituting (3.22) into the momentum equation in place of the standard hydrostatic term yields the modified flux vector

$$\underline{F}(\underline{U}) = \begin{bmatrix} q \\ \dfrac{q^2}{h} + gh \log\left(\dfrac{h}{h_{\mathrm{ref}}}\right) \end{bmatrix} \tag{3.23}$$

while the mass conservation equation and the state vector $\underline{U} = [h, q]^T$ remain unchanged. The system retains the conservative form (3.2) with the flux (3.23), and the same FVM update formula (3.9) applies without modification: only the evaluation of the flux vector at each interface changes. The mathematical properties of the modified system differ from those of the standard SWE because the Jacobian $\mathbf{J} = \partial \underline{F} / \partial \underline{U}$ now takes the form

$$\mathbf{J}(\underline{U}) = \begin{bmatrix} 0 & 1 \\ -\dfrac{q^2}{h^2} + g \log\left(\dfrac{h}{h_{\mathrm{ref}}}\right) + g & \dfrac{2q}{h} \end{bmatrix} \tag{3.24}$$

The eigenvalues of $\mathbf{J}$, which govern the characteristic speeds of the system, are

$$\lambda_{1,2} = u \mp c_{\text{mod}}, \qquad c_{\text{mod}} = \sqrt{g\left[\log\left(\frac{h}{h_{\text{ref}}}\right) + 1\right]} \tag{3.25}$$

where c_{mod} plays the role of the wave celerity for the modified system. Comparing with the standard celerity $c = \sqrt{gh}$, the modified celerity c_{mod} depends logarithmically rather than algebraically on h, which has several physical consequences. For $h > h_{\text{ref}}e^{-1}$ the argument of the square root in (3.25) is positive and the system remains hyperbolic; below this threshold the characteristic speeds become complex, signaling a loss of hyperbolicity that renders the initial-value problem ill-posed.

Critical remark on the dam-break test In the dam-break initial data (case a, section 3.2.1), the right state has $h_R = 0.1\,\text{m}$. Since

$$h_R = 0.1 < h_{\text{ref}}e^{-1} = e^{-1} \approx 0.368\,\text{m} \tag{3.26}$$

the modified system (3.23) is ill-posed ($c_{\text{mod}}^2 < 0$) in the right half of the domain at $t = 0$. In the implementation, the modified flux falls back to the Rusanov solver, in which the celerity is clamped to a small positive value wherever the radicand in (3.25) becomes negative; the associated numerical diffusion artificially stabilizes what is in fact an ill-posed problem, so the computed solution there lacks physical justification. The results of figure 3.4 for the modified-pressure dam break must therefore be interpreted with caution: they are numerically stable but not physically meaningful in the region $h < h_{\text{ref}}e^{-1}$.

3.3.2 Numerical Results and Discussion

The modified flux (3.23) is tested with $h_{\text{ref}} = 1$ on the dam-break and stationary-shock initial data. For the standard system the exact Riemann solver of section 3.1.5 is used, whereas for the modified system no practical exact solver is available: the interface flux is therefore evaluated with the Rusanov (local Lax-Friedrichs) approximate solver

$$\underline{F}^{\text{Rus}}_{i+1/2} = \frac{1}{2}\left[\underline{F}(\underline{U}^n_i) + \underline{F}(\underline{U}^n_{i+1})\right] - \frac{S_{i+1/2}}{2}\left(\underline{U}^n_{i+1} - \underline{U}^n_i\right) \tag{3.27}$$

$$S_{i+1/2} = \max\left(|u^n_i| + c^n_i, |u^n_{i+1}| + c^n_{i+1}\right) \tag{3.28}$$

where the dissipation term proportional to the maximum local wave speed $S_{i+1/2}$ penalizes the jump across the interface. To isolate the effect of the pressure law, the standard system is shown with its first-order Godunov solution; both schemes are first-order, so any difference between the two curves reflects the pressure law alone rather than scheme-dependent dissipation or dispersion.

Dam break with modified pressure Figure 3.4 compares the two Godunov solutions at time $T = 1\,\text{s}$ for the dam-break initial data ($h_L = 1$, $h_R = 0.1$). The most immediate effect of the modified pressure is a marked reduction in the rightward shock speed: at $T = 1$ the shock front computed with the standard flux has already reached and reflected from the right wall, whereas the modified-pressure shock is still propagating through the interior of the domain, having traveled a substantially shorter distance. This behavior is in direct agreement with the eigenvalue analysis of section 3.3.1: for $h \leq 1$ and $h_{\text{ref}} = 1$ one has $\log(h/h_{\text{ref}}) \leq 0$, so that $c_{\text{mod}} \leq \sqrt{g}$ is strictly smaller than the corresponding standard celerity $c = \sqrt{gh}$ evaluated at the same depth. The weaker pressure gradient of the logarithmic law provides less driving force per unit height, resulting in a slower propagation of the shock front and a less pronounced post-shock height plateau. The rarefaction fan on the left is also affected: the modified solution shows a smoother and more extended depletion profile rather than the flat constant-state region produced by the standard flux, reflecting the altered characteristic fan structure. The discharge panel confirms both observations: the peak value of q is significantly lower in the modified case, and the profile shape is qualitatively different, with a more gradual spatial variation across the entire domain.

Stationary shock with modified pressure The stationary-shock data (case c) are revisited here with the modified pressure law, shown in figure 3.5. As established in section 3.2.2, the zero-momentum initial data $(h_L, h_R, q) = (1, 0.5, 0)$ do not produce a stationary shock even in the standard system: the pressure imbalance drives a finite shock speed in both cases. The relevant comparison is therefore between the two shock speeds. Replacing the standard flux with the modified flux (3.23) alters the second component of the flux vector at every height level, so that the same left and right states produce a different flux jump $[\underline{F}_{\text{mod}}] \neq [\underline{F}_{\text{std}}]$ and hence a different speed. The Rankine-Hugoniot condition for the modified system reads

$$\sigma[\underline{U}] = [\underline{F}_{\text{mod}}(\underline{U})] \tag{3.29}$$

A simple way to compare the two shock speeds is to combine the two scalar components of the Rankine-Hugoniot condition and eliminate $[q]$, which gives the heuristic estimate $\sigma^2 \approx [\underline{F}_{\cdot,2}]/[h]$ evaluated on the initial states. This is not the exact shock speed (the first jump condition with $q_L = q_R = 0$ would force $\sigma[h] = 0$, so the true speed must be obtained from the full Riemann solution), but it is adequate to compare the two pressure laws on equal footing:

$$\sigma_{\text{std}}^2 \approx \frac{\frac{1}{2}g[h^2]}{[h]} = \frac{1}{2}g(h_L + h_R) \quad \Longrightarrow \quad \sigma_{\text{std}} \approx 2.71\,\text{m/s} \tag{3.30}$$

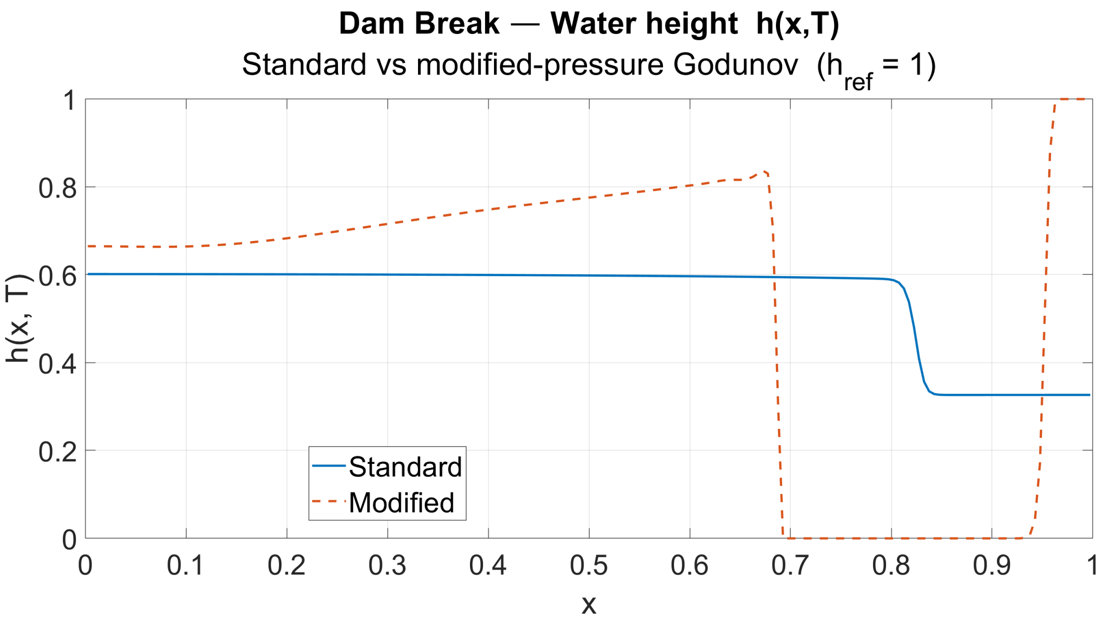
$$\sigma_{\text{mod}}^2 \approx \frac{g\left[h\log\left(\frac{h}{h_{\text{ref}}}\right)\right]}{[h]} = \frac{9.81 \times [0.5\log(0.5) - \log(1)]}{-0.5} \approx 6.80\,\text{m}^2/\text{s}^2 \quad \Longrightarrow \tag{3.31}$$

$$\Longrightarrow \quad \sigma_{\text{mod}} \approx 2.61\,\text{m}^2/\text{s}^2 \tag{3.32}$$

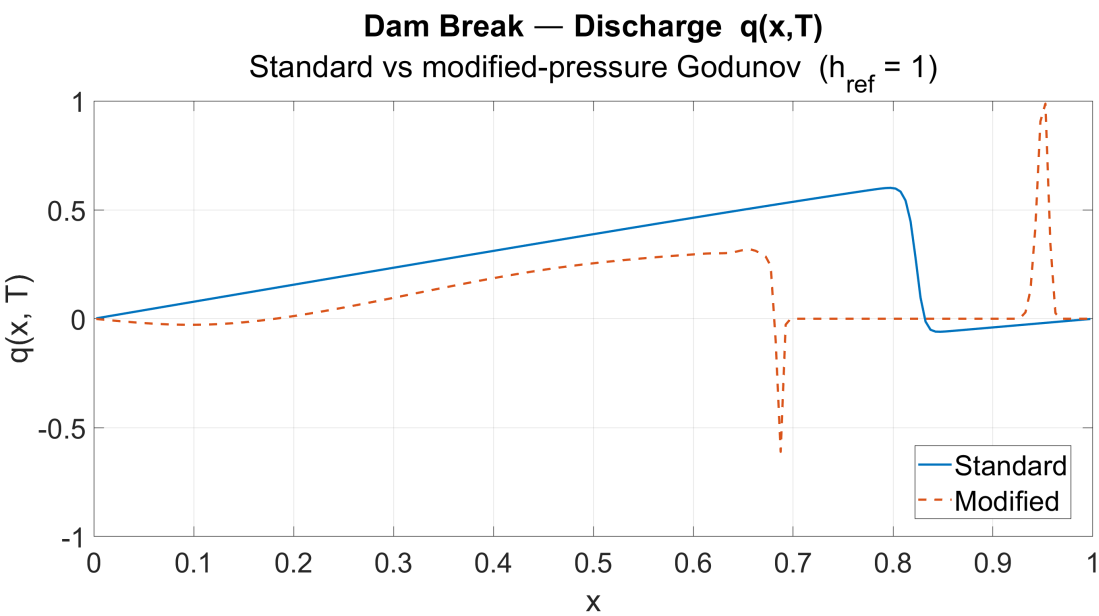
The logarithmic pressure law produces a weaker shock speed than the standard SWE for the same initial height jump, consistent with the eigenvalue comparison $c_{\text{mod}} < c$ established in section 3.3.1. Both speeds are strictly positive, confirming that neither configuration is stationary; the modified shock is the slower of the two. This is precisely what figure 3.5 shows: at $T = 1\,\text{s}$ the two fronts sit at different positions, and the modified solution carries its momentum with a different, spatially shifted discharge profile. Both solutions retain the sharp, monotone profile characteristic of the first-order schemes, and the pre-shock heights on the left plateau differ slightly, reflecting the different intermediate states produced by the two flux functions.

3.3.3 The Need for Slope Limiters

The three Riemann tests of section 3.2 expose the fundamental limitation of the pure Lax-Wendroff reconstruction. In every case the unlimited second-order scheme lost stability within a few hundredths of a second: the dam break and the reverse dam break broke down at $t \approx 0.011\,\text{s}$, and the stationary-shock case at $t \approx 0.026\,\text{s}$, all far short of the final time $T = 1\,\text{s}$. The first-order Godunov scheme, by contrast, remained monotone and stable



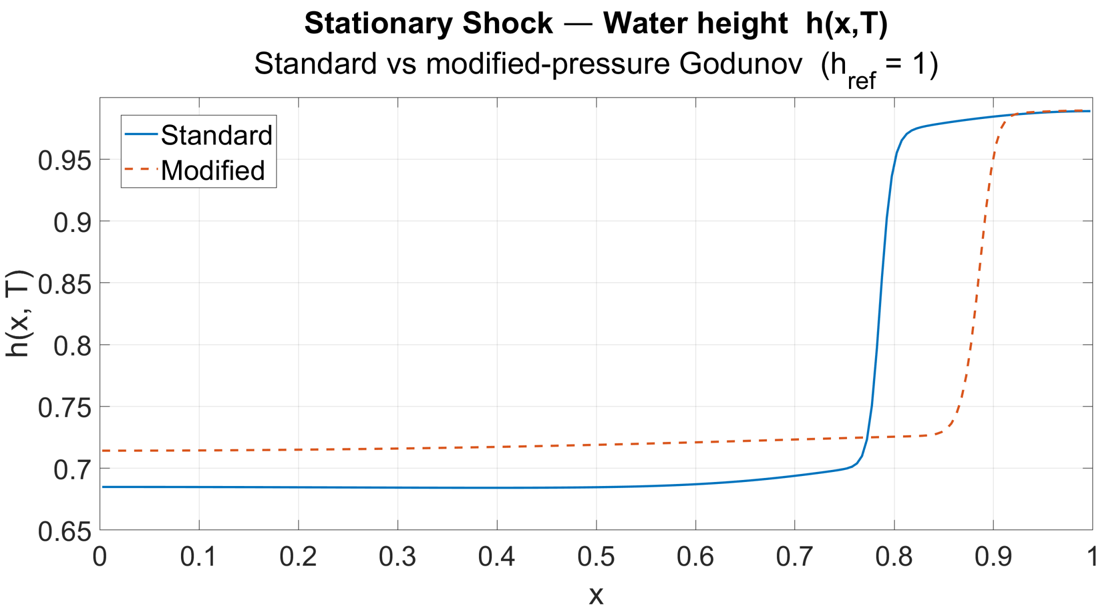
(a) Water height $h(x, T)$. The modified-pressure shock (dashed orange) travels more slowly than the standard one (solid blue), consistent with $c_{\text{mod}} \leq \sqrt{g} < c$ for $h \leq h_{\text{ref}} = 1$; the post-shock plateau is lower and the rarefaction is more gradual. In the right portion of the domain, where $h < h_{\text{ref}}e^{-1}$, the modified system is non-hyperbolic and the corresponding part of the curve is a numerical artifact of the clamped Rusanov flux, not a physical solution



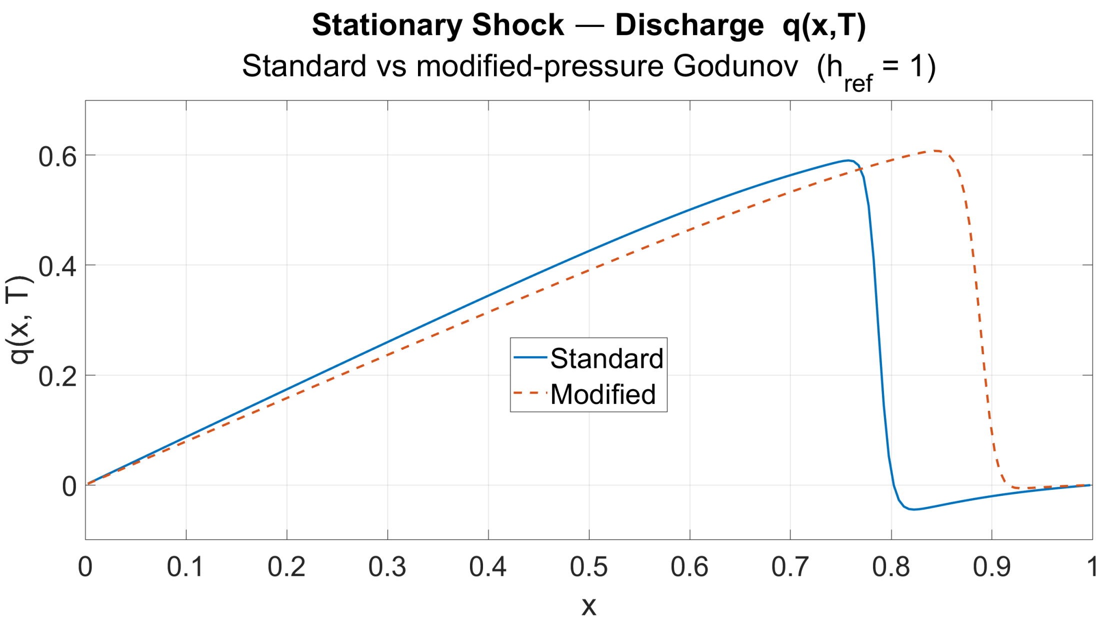
(b) Discharge $q(x, T)$. The peak discharge is substantially reduced in the modified case, reflecting the weaker momentum transfer of the logarithmic pressure term; the violent spikes near the right wall coincide with the non-hyperbolic region and have no physical meaning

Figure 3.4: Dam-break Riemann problem at $T = 1\,\text{s}$: standard Godunov (solid blue) versus modified-pressure Rusanov with $h_{\text{ref}} = 1$ (dashed orange). The modified shock is slower, but the solution is physically meaningful only where $h > h_{\text{ref}}e^{-1} \approx 0.368$ (see the remark in section 3.3.1). Physical parameters: $L = 1\,\text{m}$, $g = 9.81\,\text{m/s}^2$, $h_L = 1$, $h_R = 0.1$, $h_{\text{ref}} = 1$

throughout every simulation. This contrast is not a coincidence but a direct consequence of Godunov's theorem, which states that no linear scheme of order higher than one can be monotone: the linear reconstruction (3.13)–(3.14) is necessarily unbounded near a discontinuity, where it overshoots the neighboring cell averages and drives the solution out of the physically admissible range. The standard remedy is to make the reconstruction nonlinear



(a) Water height $h(x,T)$. Both fronts are non-stationary; the modified-pressure front (dashed orange) sits at a different position from the standard one (solid blue), consistent with the slower modified shock speed estimated in (3.32). The pre-shock plateaus differ slightly, reflecting the different intermediate states of the two flux functions



(b) Discharge $q(x,T)$. The modified solution carries its momentum with a spatially shifted profile, confirming that both configurations are actively evolving rather than force-balanced

Figure 3.5: Stationary-shock data (case c) at $T = 1\,\mathrm{s}$: standard Godunov (solid blue) versus modified-pressure Rusanov with $h_{\text{ref}} = 1$ (dashed orange). Contrary to the name of the test case, neither configuration is stationary (section 3.2.2); the modified pressure law yields a different, slower shock speed, so the two fronts separate over time. Physical parameters: $L = 1\,\mathrm{m}$, $g = 9.81\,\mathrm{m/s^2}$, $h_L = 1$, $h_R = 0.5$, $h_{\text{ref}} = 1$

through a slope limiter. Instead of using the central slope $\frac{1}{2}(\underline{U}_{i+1} - \underline{U}_{i-1})$ unconditionally, the limited slope is reduced, or set to zero, wherever neighboring gradients change sign or grow too steep, so that no new extrema are created. The minmod limiter selects the smaller of the two one-sided slopes when they share a sign and returns zero otherwise, while the van Leer limiter applies a smooth nonlinear blending of the same slopes. In both cases the scheme retains second-order accuracy in smooth regions, where the limiter stays inactive, but degrades gracefully to the monotone first-order behavior at discontinuities. The result is a Total Variation Diminishing (TVD) scheme that combines the sharp resolution of Lax-Wendroff in smooth regions with the robustness of Godunov at shocks. A complementary observation concerns the time integration. The reconstruction used here is paired with the forward-Euler update (3.9), which is not strong-stability-preserving (SSP): even with a slope limiter, a single forward-Euler step can amplify oscillations that the spatial limiter is meant to suppress. A fully robust second-order scheme therefore combines a TVD slope limiter in space with an SSP time integrator, such as the two-stage SSP Runge-Kutta (Heun) method, or equivalently a MUSCL-Hancock predictor-corrector update. The implementation of either remedy is left as a natural extension of the present work; the comparison carried out here is sufficient to establish, on physical and numerical grounds, why such measures are indispensable for second-order finite-volume schemes applied to discontinuous solutions.